



Pokerbots 2026

Lecture 12: Noam Brown

Sponsors



CITADEL | CITADEL Securities



Announcements

Tuesday 1/27

Team Strategy Reports due **YESTERDAY**

Submit ASAP if you have not already!!!

Wednesday 1/28 (TODAY)

Final Bot Submission due 11:59PM EST

- Upload and select bot as active on scrimmage server
- Both report and bot needed to pass this class!
- Bot will compete in last and final Pokerbots tournament
- Non-secret prize amounts listed on syllabus

Final Tournament Prizes	
First place	\$10,000
Second place	\$6,500
Third place	\$3,500
Fourth place	\$2,000
Fifth place	\$1,000
First place in language (Python, Java, or C++)	\$500 x 3
Second place in language (Python, Java, or C++)	\$250 x 3
Third place in language (Python, Java, or C++)	\$125 x 3
Best freshman-majority (>51%) team	\$2,000

Friday 1/30

Pokerbots Final Event 4:30-7PM in Tang Center (E51)

- Presentation of Awards
- Closing Ceremony
- Sponsor Event and Puzzle Hunt
- Lots of free merch and raffle prizes!
- Dinner provided 😊

DATE TBD (probably ~beginning of February)

Epic Poker Tournament!

- Location TBA
- \$3500 cash prize pool

A large, stylized blue flower graphic is positioned on the left side of the image. It is composed of several overlapping circles of varying shades of blue, creating a layered, petal-like effect. The flower is centered vertically and extends from the bottom edge towards the top.

Giveaway Game!

Evaluation Raffle

- MIT class evaluations have opened
 - pkr.bot/eval
- To enter today's raffle, make sure to fill out eval!
- Then enter your kerb at pkr.bot/evalraffle
 - Must have proof of completion to get prize!
- Prize: JBL Clip 5



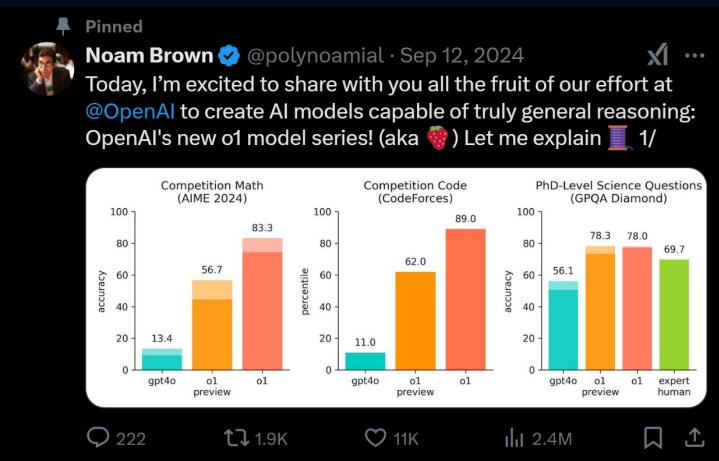


Dr. Noam Brown

Dr. Noam Brown

Guest Lecture: Dr. Noam Brown

- Won Pokerbots and co-created Deep CFR while PhD at CMU
- Created Libratus and Pluribus, the world's first superhuman poker AIs
- Research scientist currently at OpenAI, previously worked at Meta AI
- Leading role in developing OpenAI's latest GPT o1 and o3 LLMs



Building a Superhuman AI for No-Limit Poker

Noam Brown

Facebook AI Research

“And that’s why there’s never going to be a computer that will play World Class Poker. It’s a people game.”

-Doyle Brunson, *Super/System* 1979



“The analysis of a more realistic poker game than our very simple model should be quite an interesting affair.”

-John Forbes Nash, 1951

Who is the better poker player?

Option 1: Someone who, over a large enough sample size, wins head-to-head vs. any other player

Option 2: Someone who makes more money playing poker than anyone else



Who is the better poker player?

Minimax Equilibrium

Option 1: Someone who, over a large enough sample size, wins head-to-head vs. any other player

Population Best Response

Option 2: Someone who makes more money playing poker than anyone else



Minimax Equilibrium

Minimax Equilibrium: a set of strategies in which no player can improve by deviating

In two-player zero-sum games, playing a minimax equilibrium ensures you will not lose in expectation

Exploitability: How much we'd lose to a best response

	Round 1	Round 2	Round 3
Us			
Best Response			

Our Exploitability = 1

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In two-player zero-sum games, playing a minimax equilibrium ensures you will not lose in expectation

Exploitability: How much we'd lose to a best response

Critical assumption: Our strategy is common knowledge, but the outcomes of random processes are **not** common knowledge

Minimax Equilibrium

Minimax Equilibrium: a set of strategies in which no player can improve by deviating

In two-player zero-sum games, playing a minimax equilibrium ensures you will not lose in expectation

Exploitability: How much we'd lose to a best response

	Round 1	Round 2	Round 3
Us			
Best Response			

Our Exploitability = 0

Minimax Equilibrium

“Poker is simple, as your opponents make mistakes, you profit.”

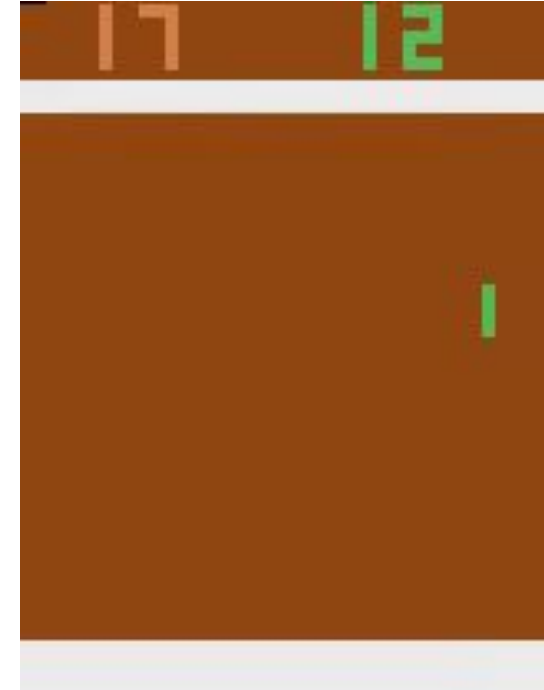
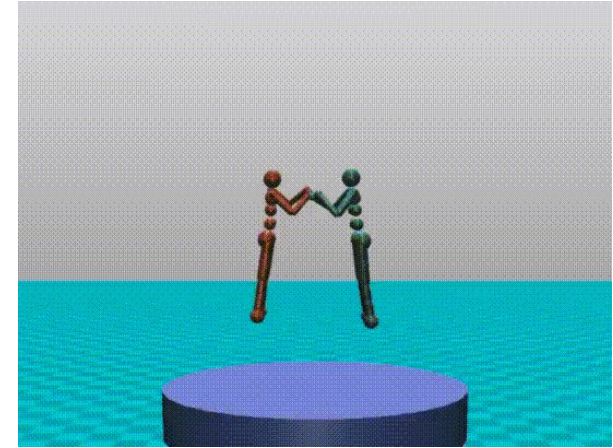
-Ryan Fee's Poker Strategy Guide

	Round 1	Round 2	Round 3
Us			
Best Response			

Our Exploitability = 0

Self-play in two-player zero-sum games

- In **self-play**, an agent gradually improves by playing against copies of itself
- Initial strategy can be completely random
- In balanced **two-player zero-sum** games, **sound self-play** provably converges to a **minimax equilibrium**
- Thus, given sufficient memory and compute, **any finite two-player zero-sum game can be “solved” via self-play**



Self-play in two-player zero-sum games

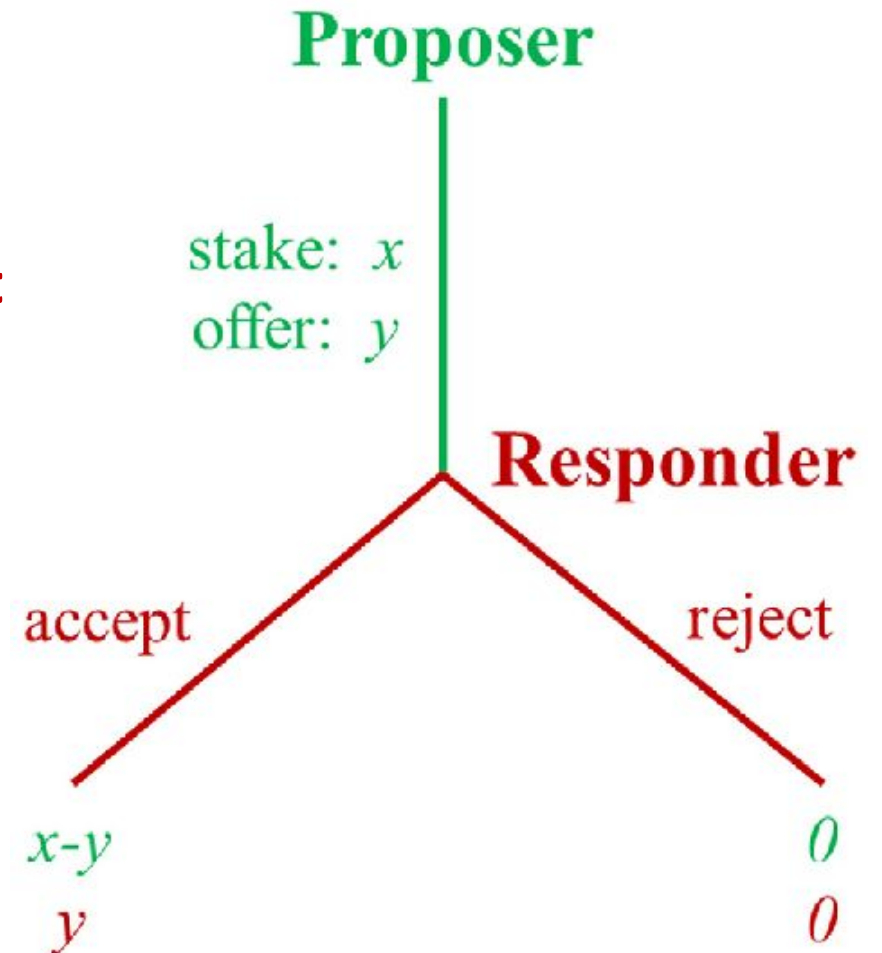
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Question: What about non-two-player zero-sum games?

Ultimatum Game

- Alice is given \$100
- First, Alice offers \$0 - \$100 to Bob
- Then, Bob must decide whether to **accept** or **reject**
 - If Bob **accepts**, then Alice and Bob keep their money
 - If Bob **rejects**, then Alice and Bob get nothing



Who is the better poker player?

~~Minimax Equilibrium~~

~~Option 1: Someone who, over a large enough sample size,
wins head-to-head vs. any other player~~

Not meaningful in general games!

Population Best Response

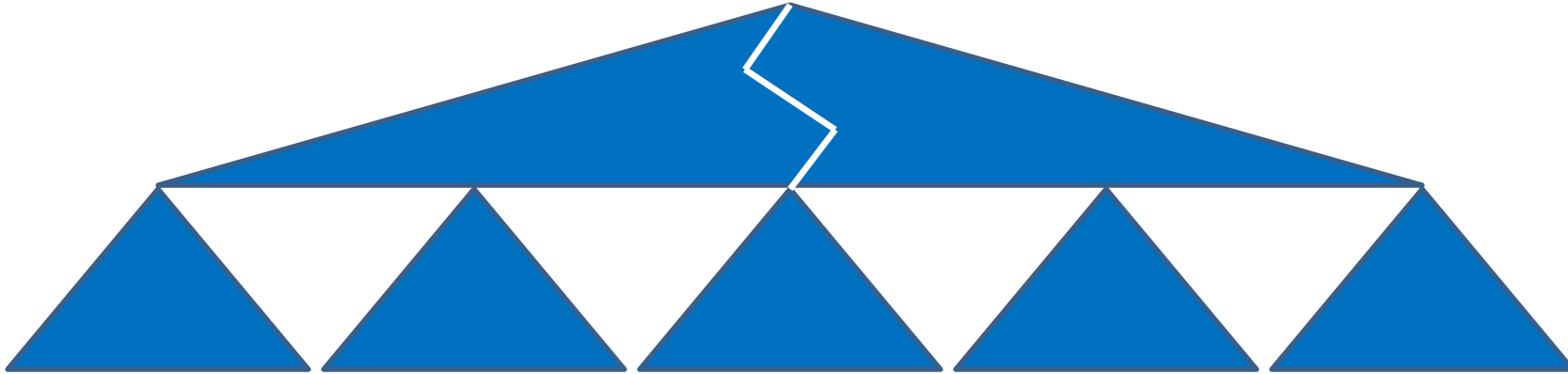
Option 2: Someone who wins more often against the
population of players than anyone else

Requires data on the population
of players, i.e., human data

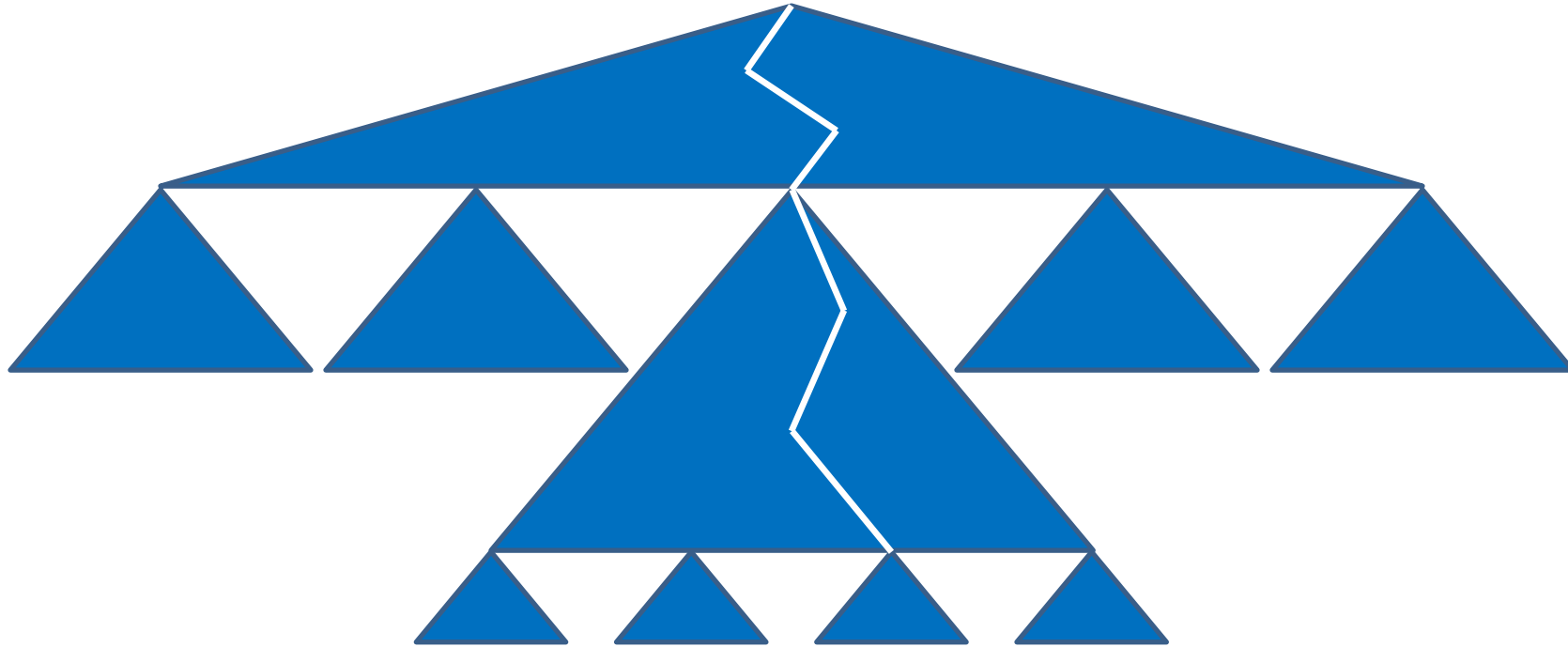
Libratus/Pluribus



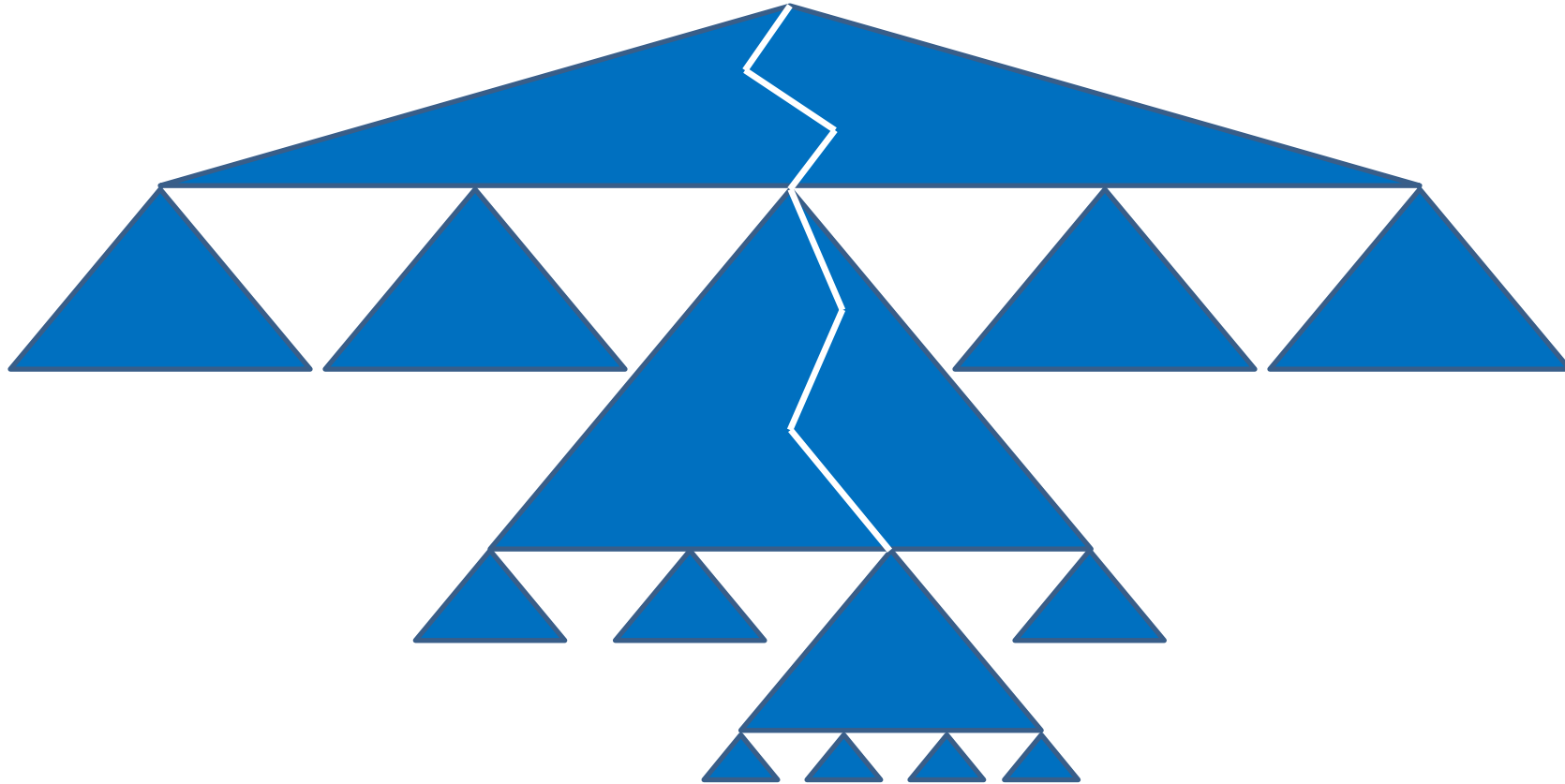
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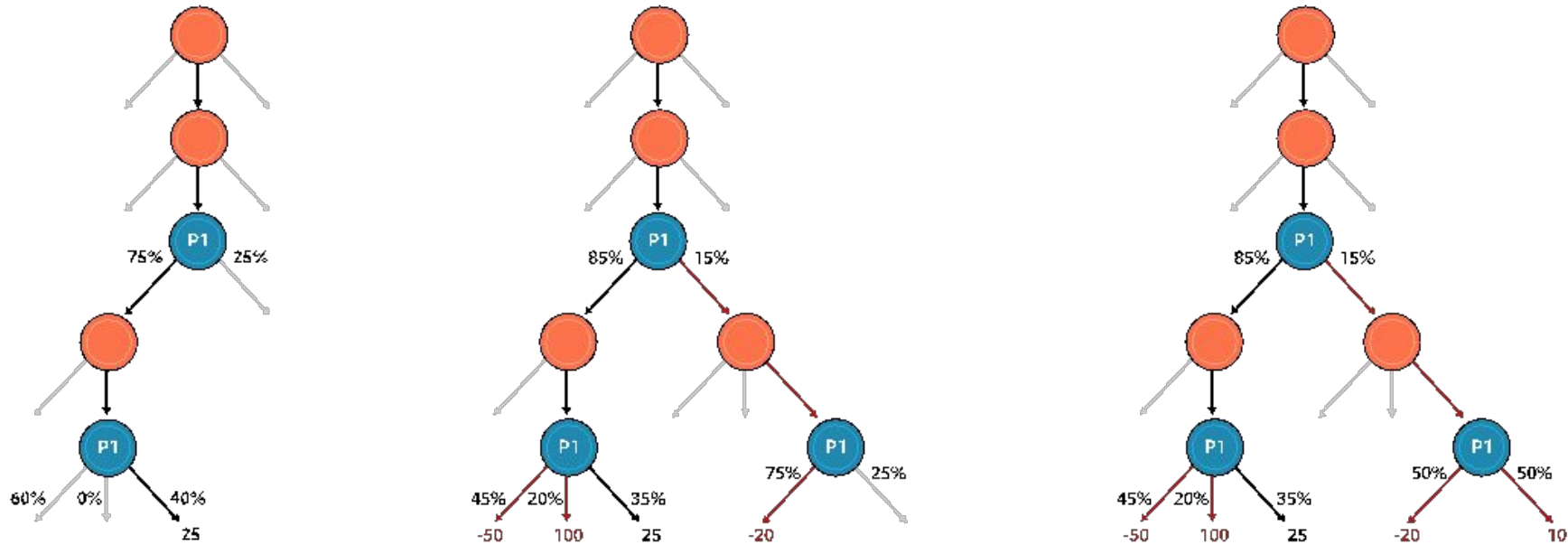
Libratus/Pluribus



Libratus/Pluribus



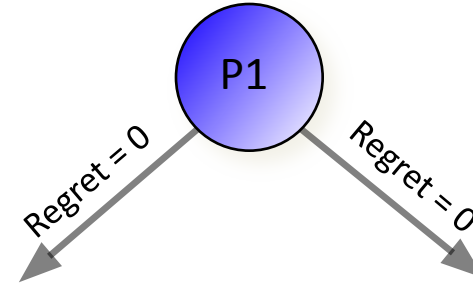
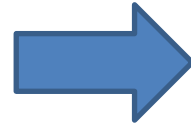
Computing Equilibria with Counterfactual Regret Minimization



Monte Carlo Counterfactual Regret Minimization (MCCFR)

[Zinkevich *et al.* NeurIPS-07, Lanctot *et al.* NeurIPS-09]

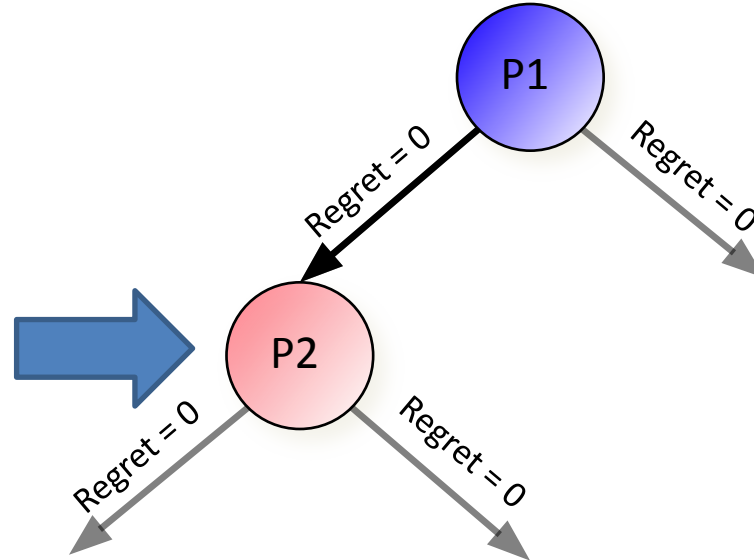
Pick action proportional to **positive** regret



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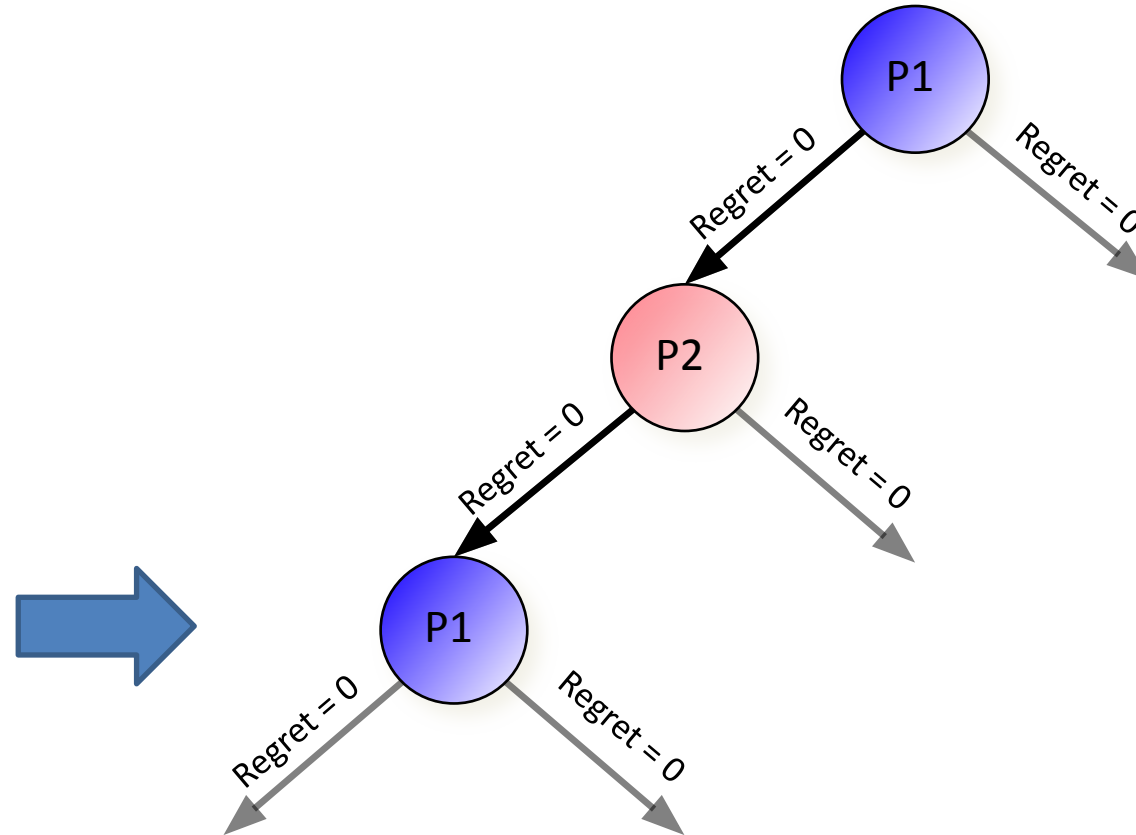
[Zinkevich *et al.* NeurIPS-07, Lanctot *et al.* NeurIPS-09]

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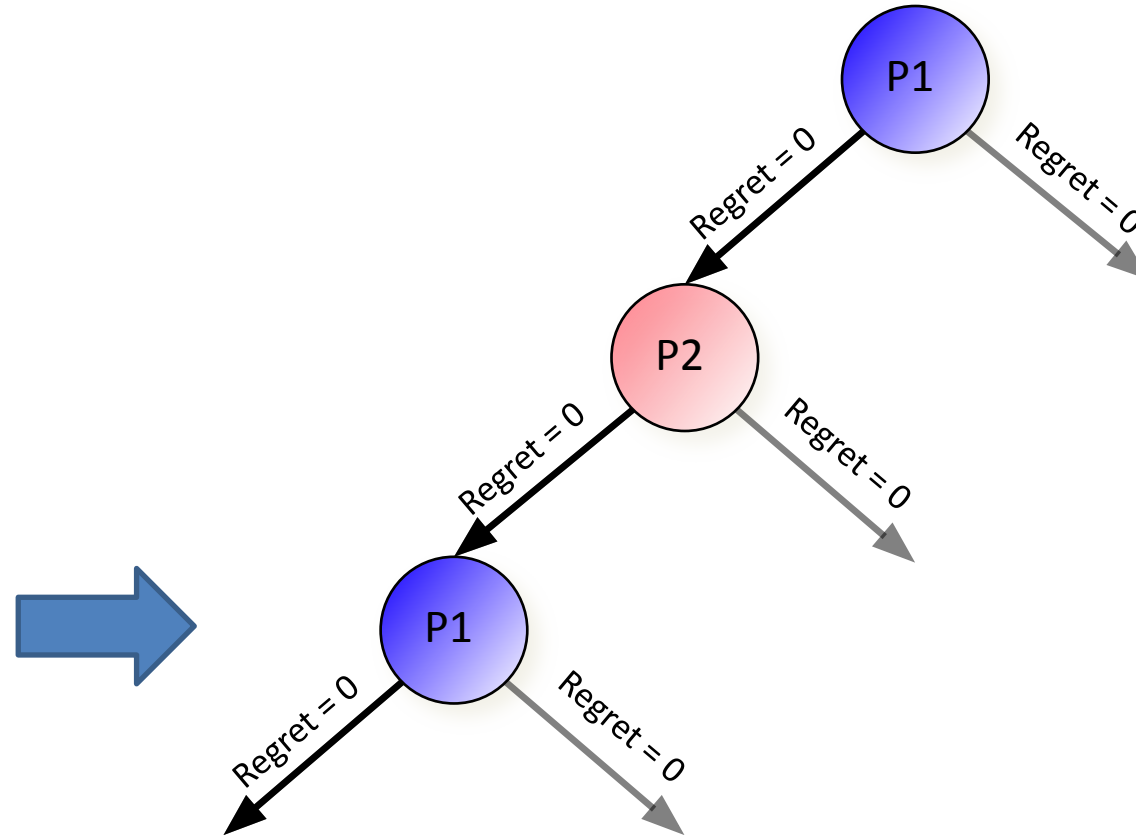
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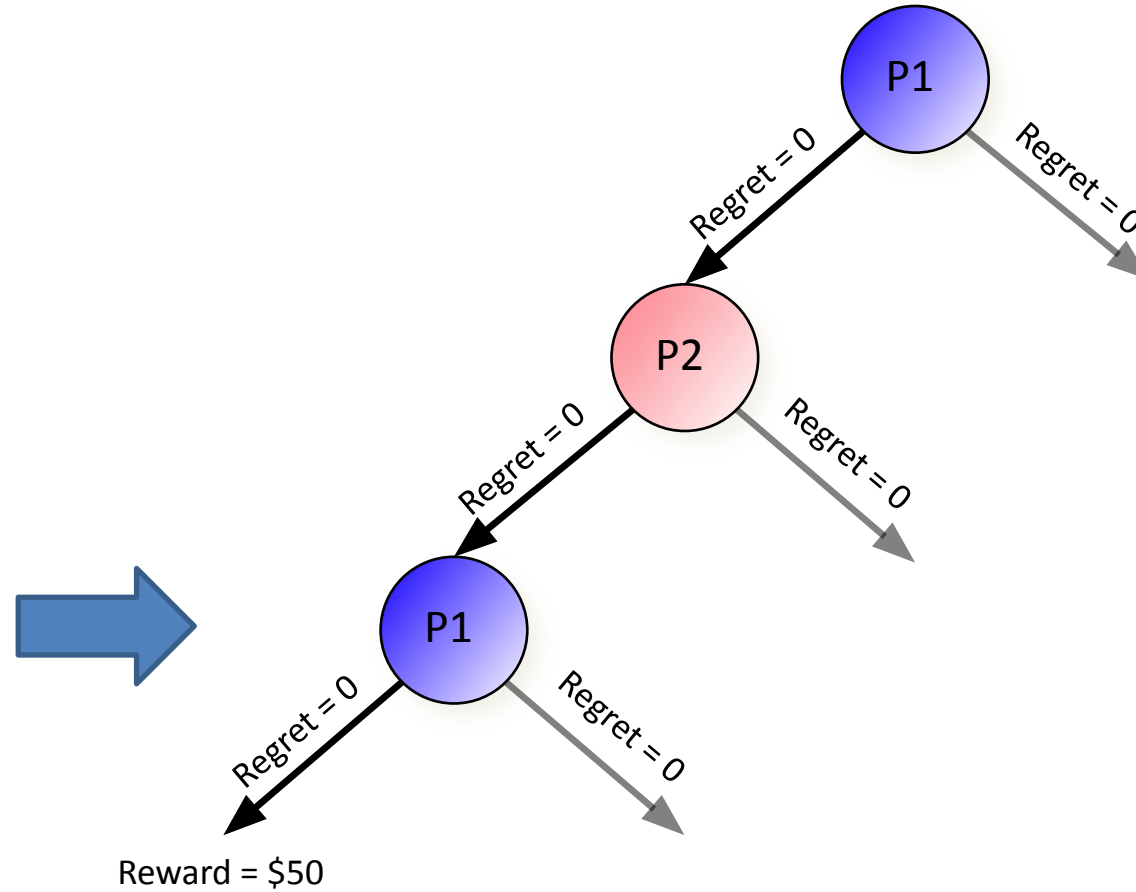
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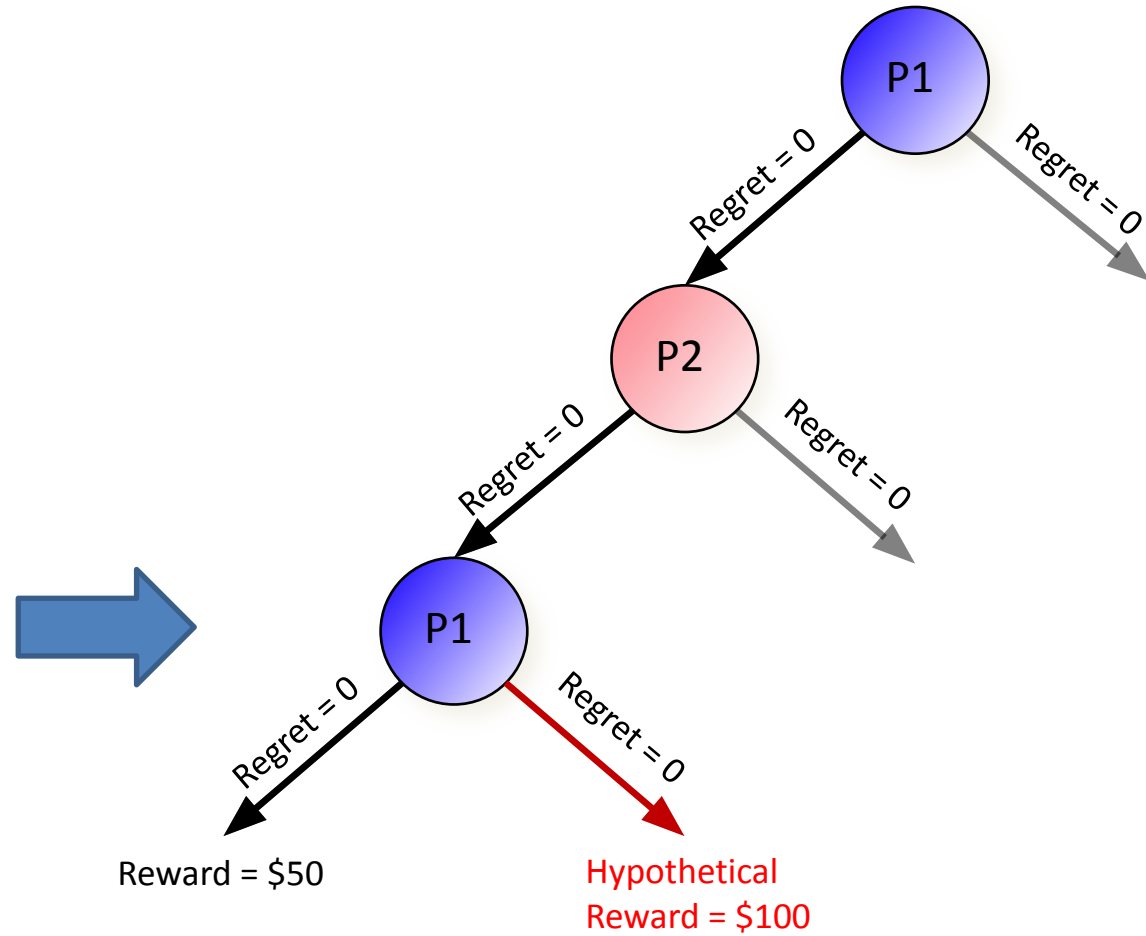
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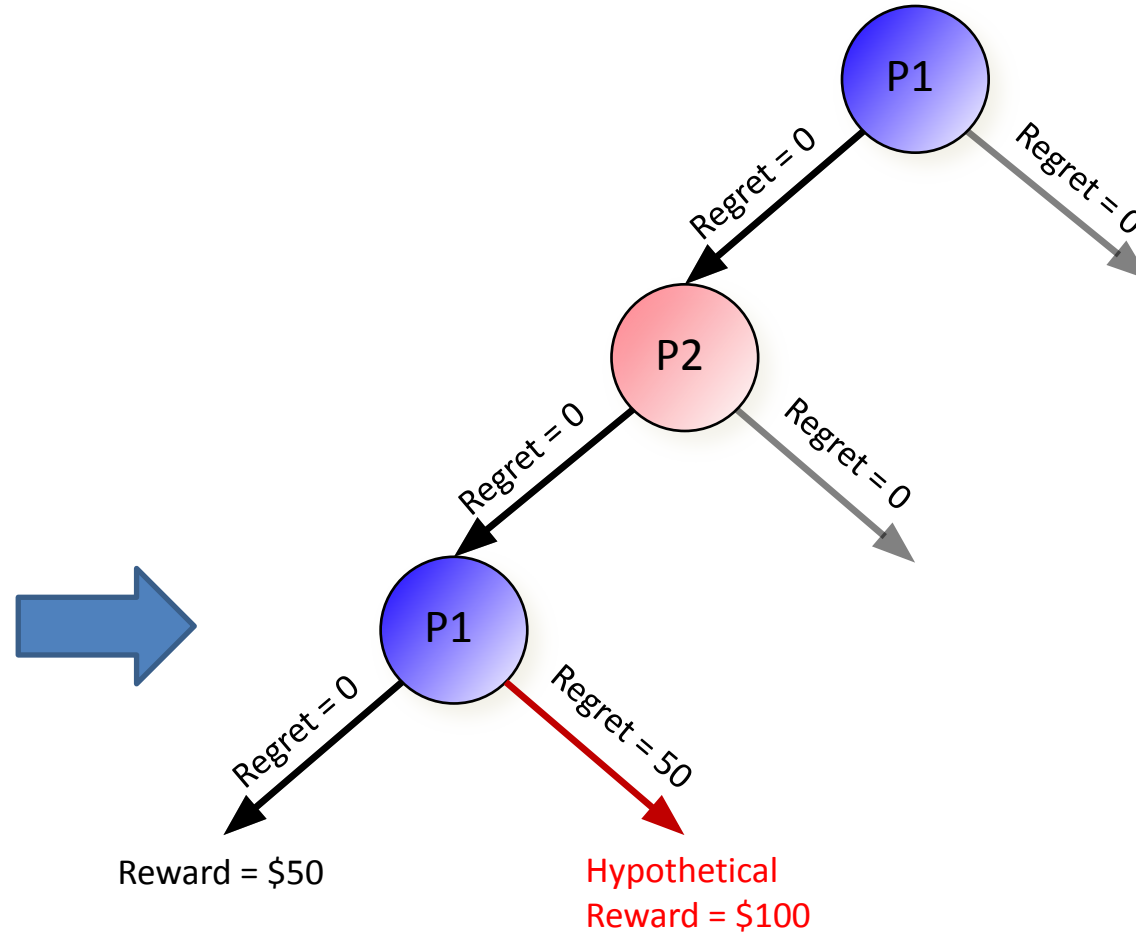
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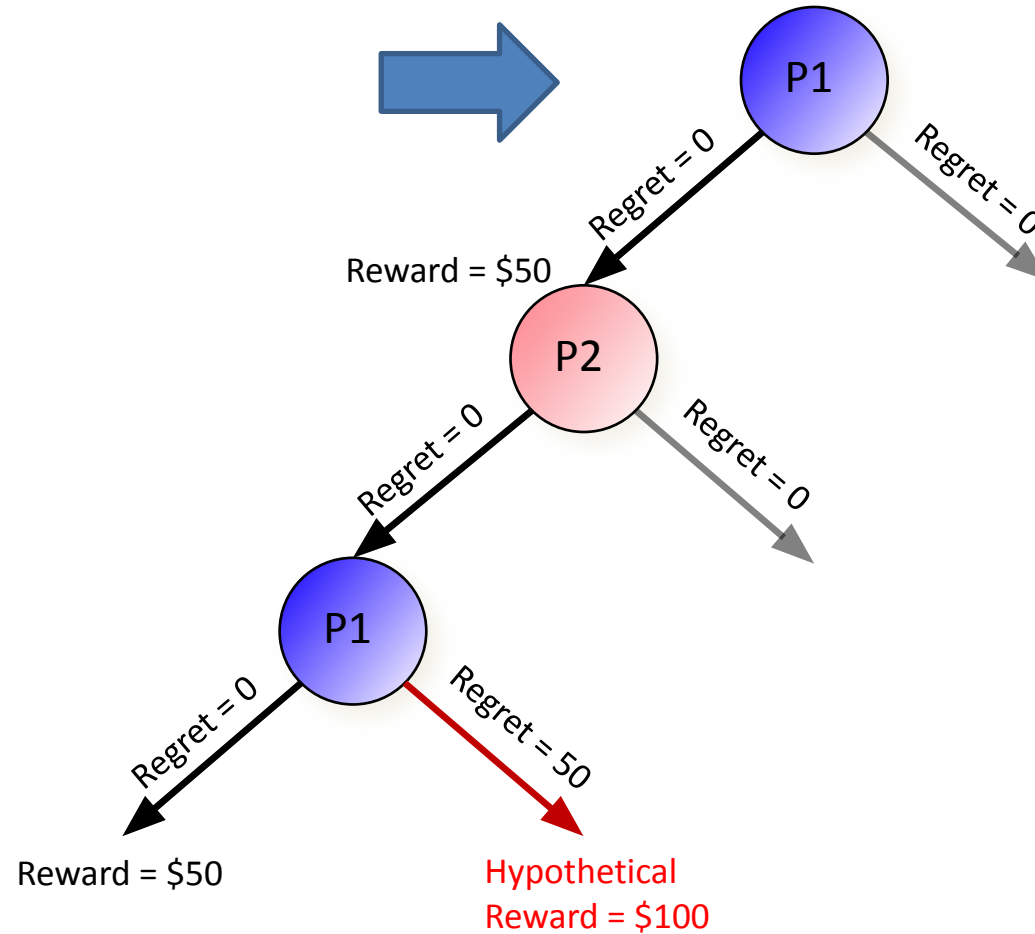
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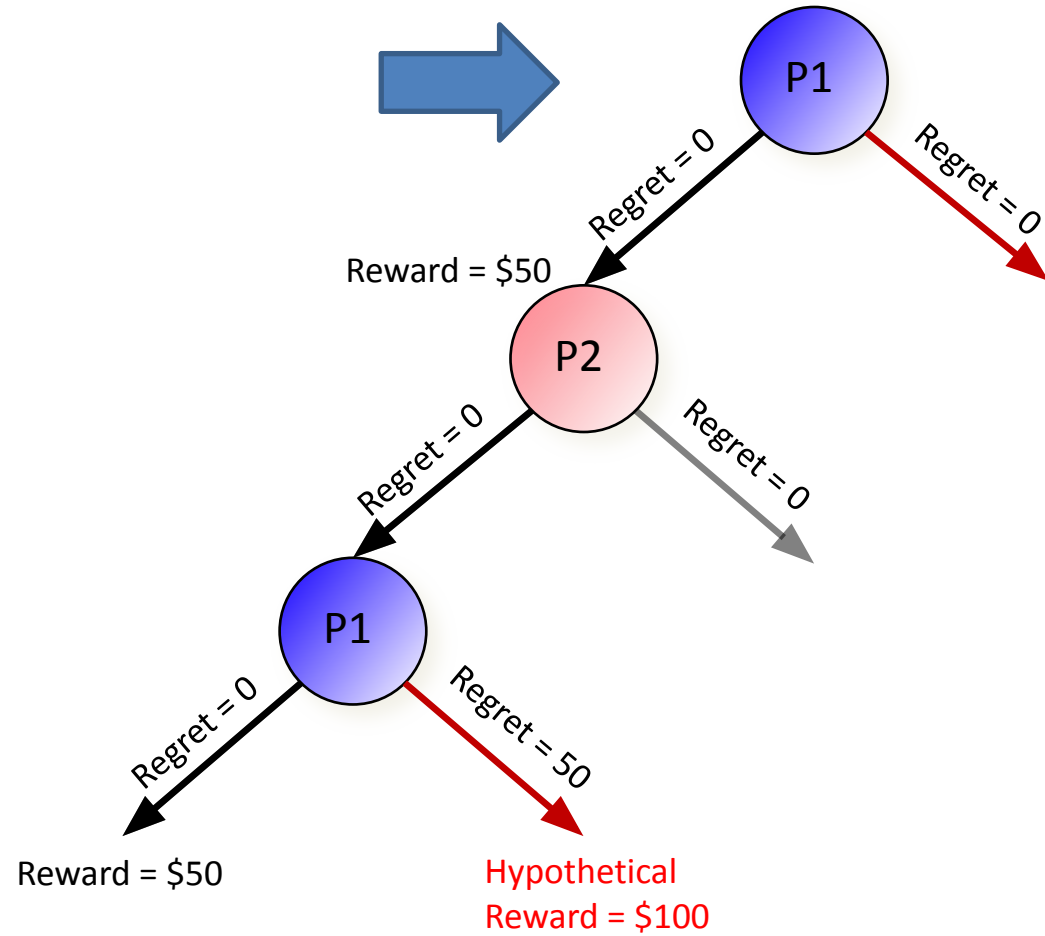
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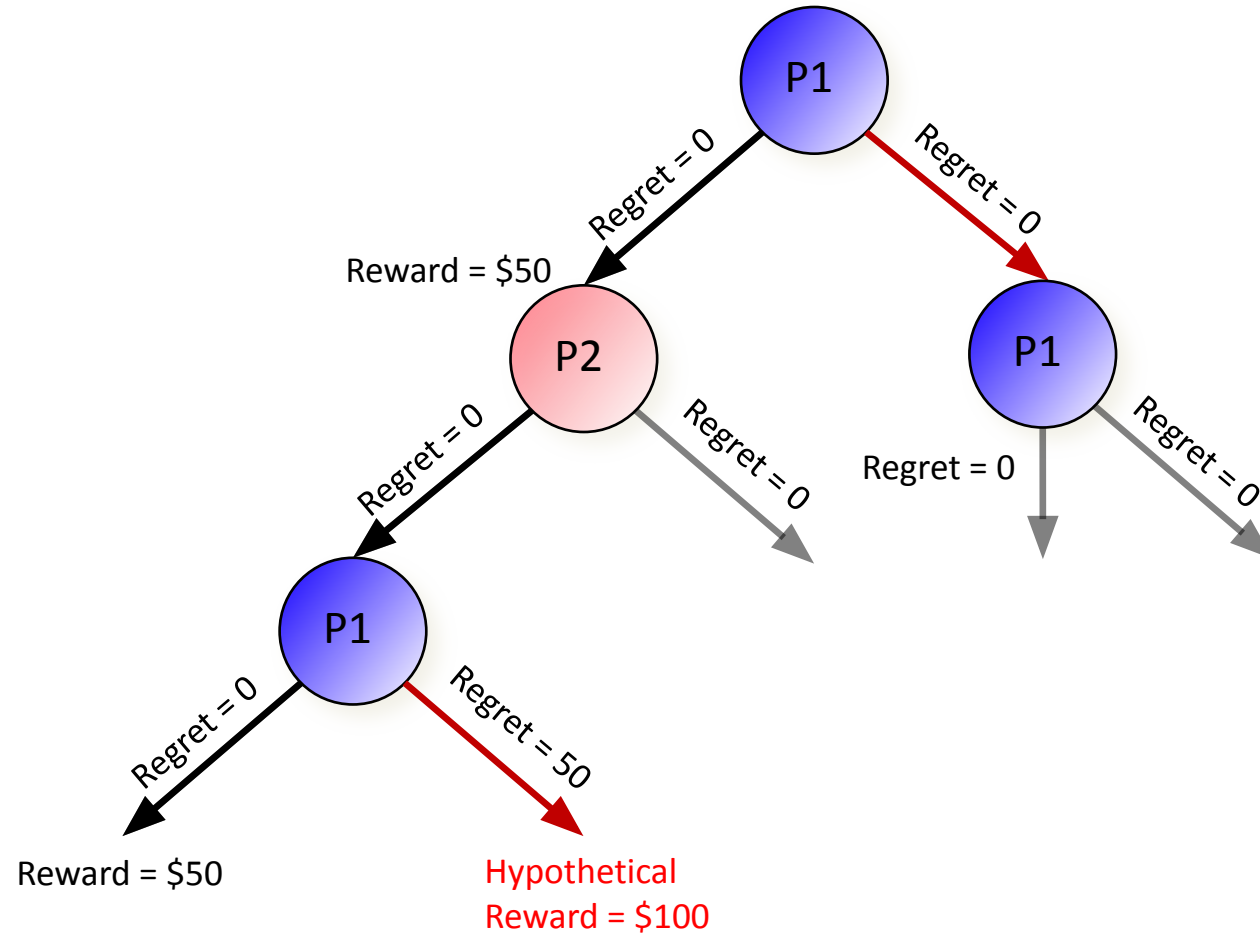
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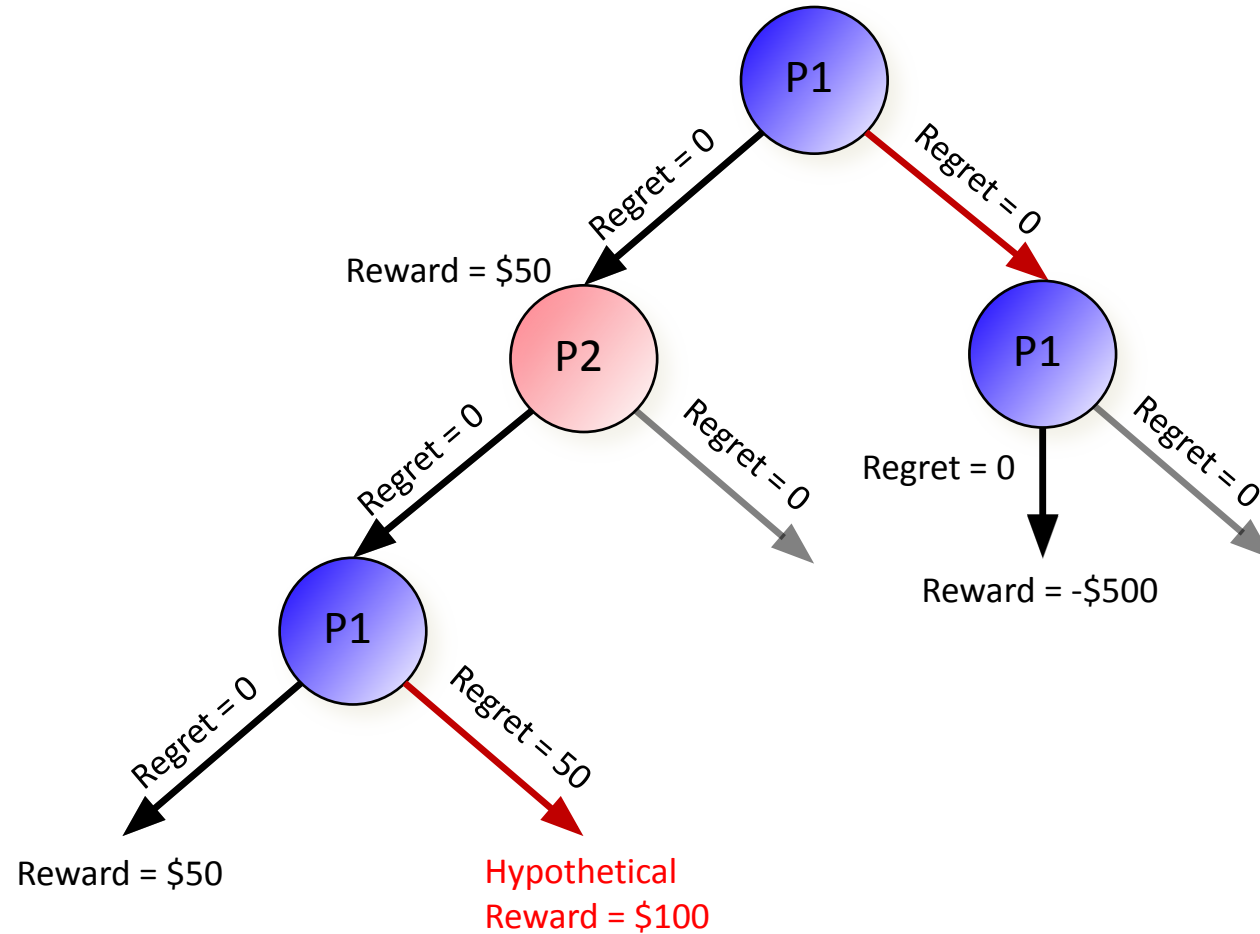
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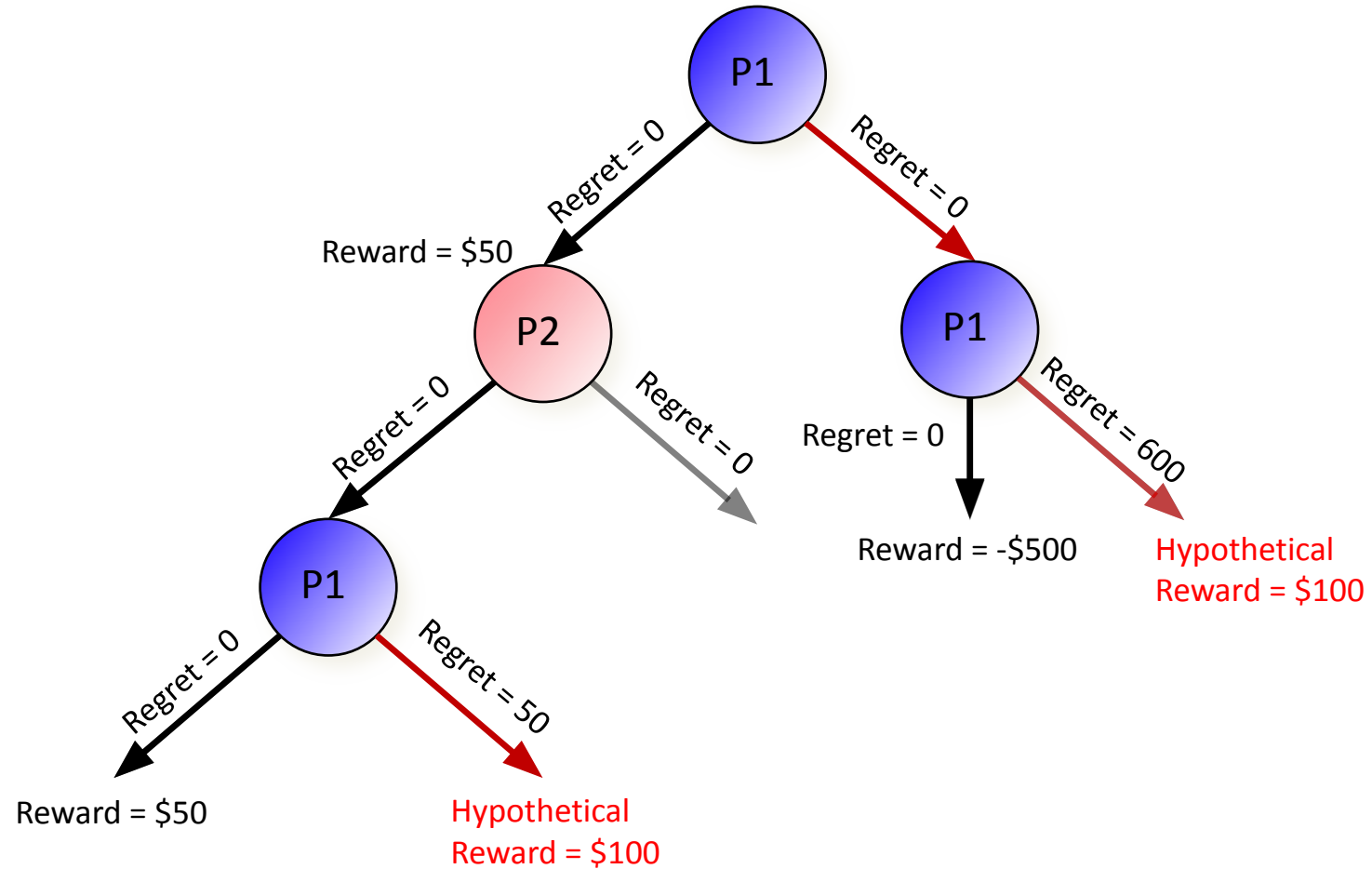
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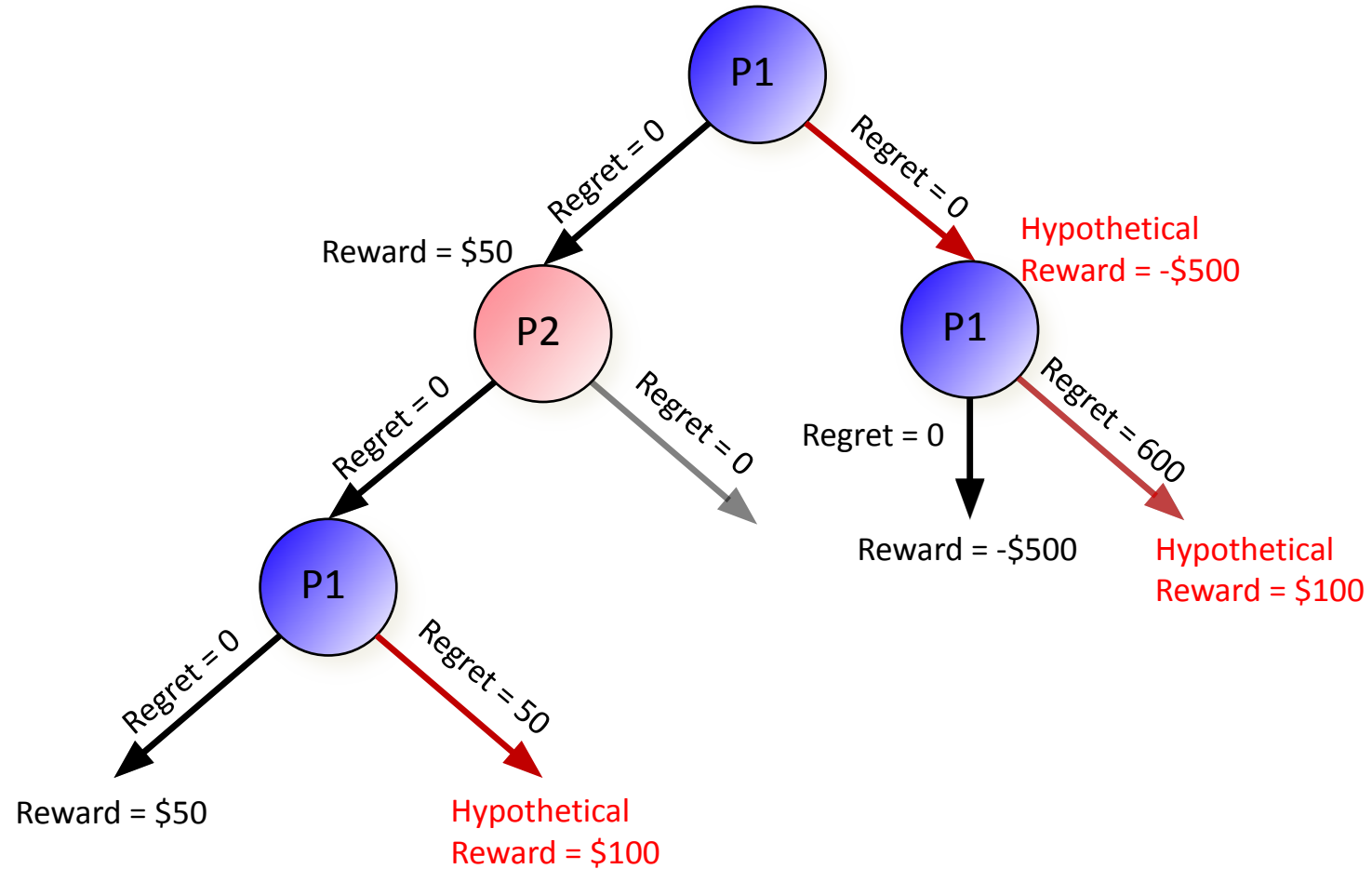
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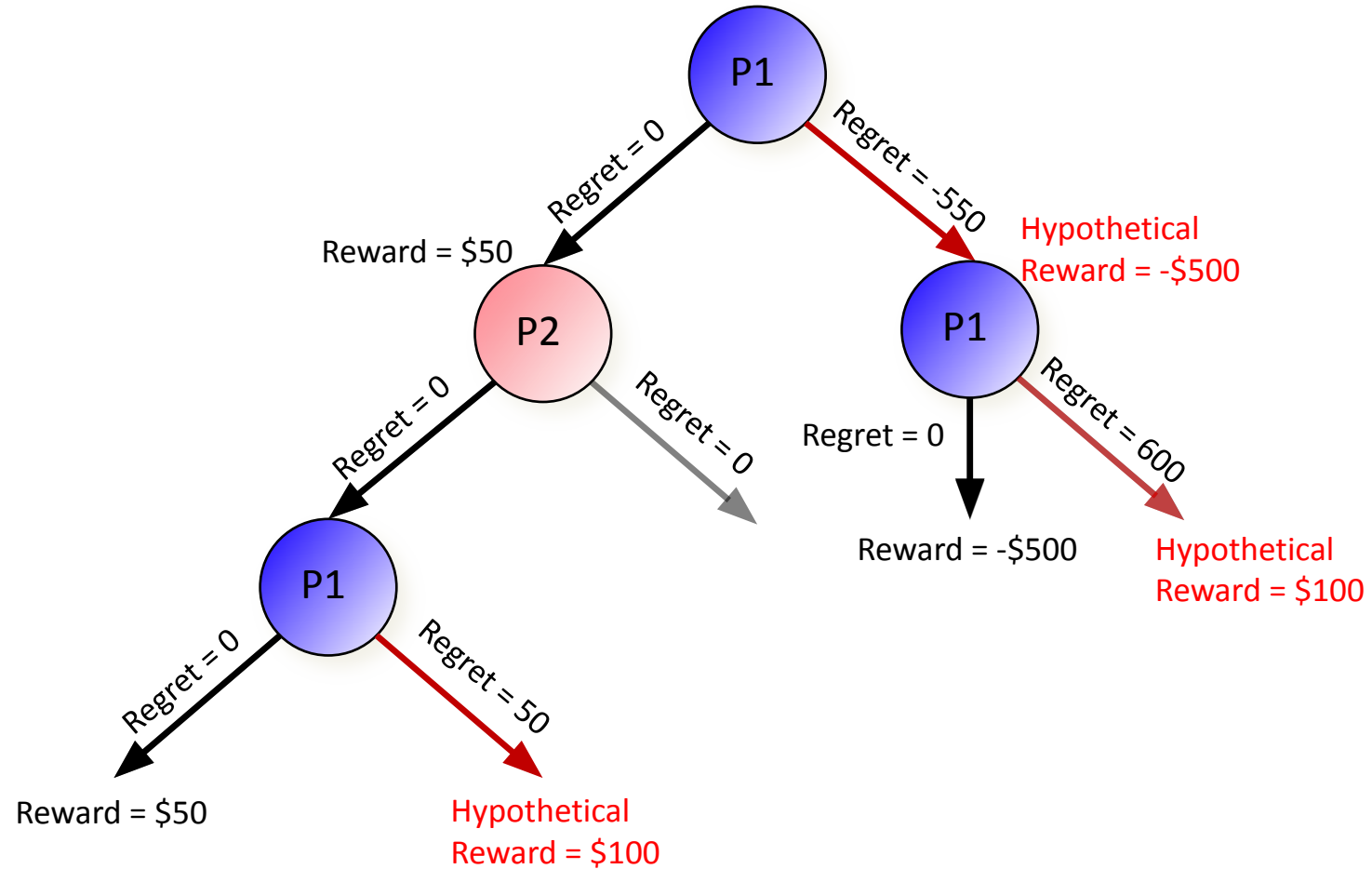
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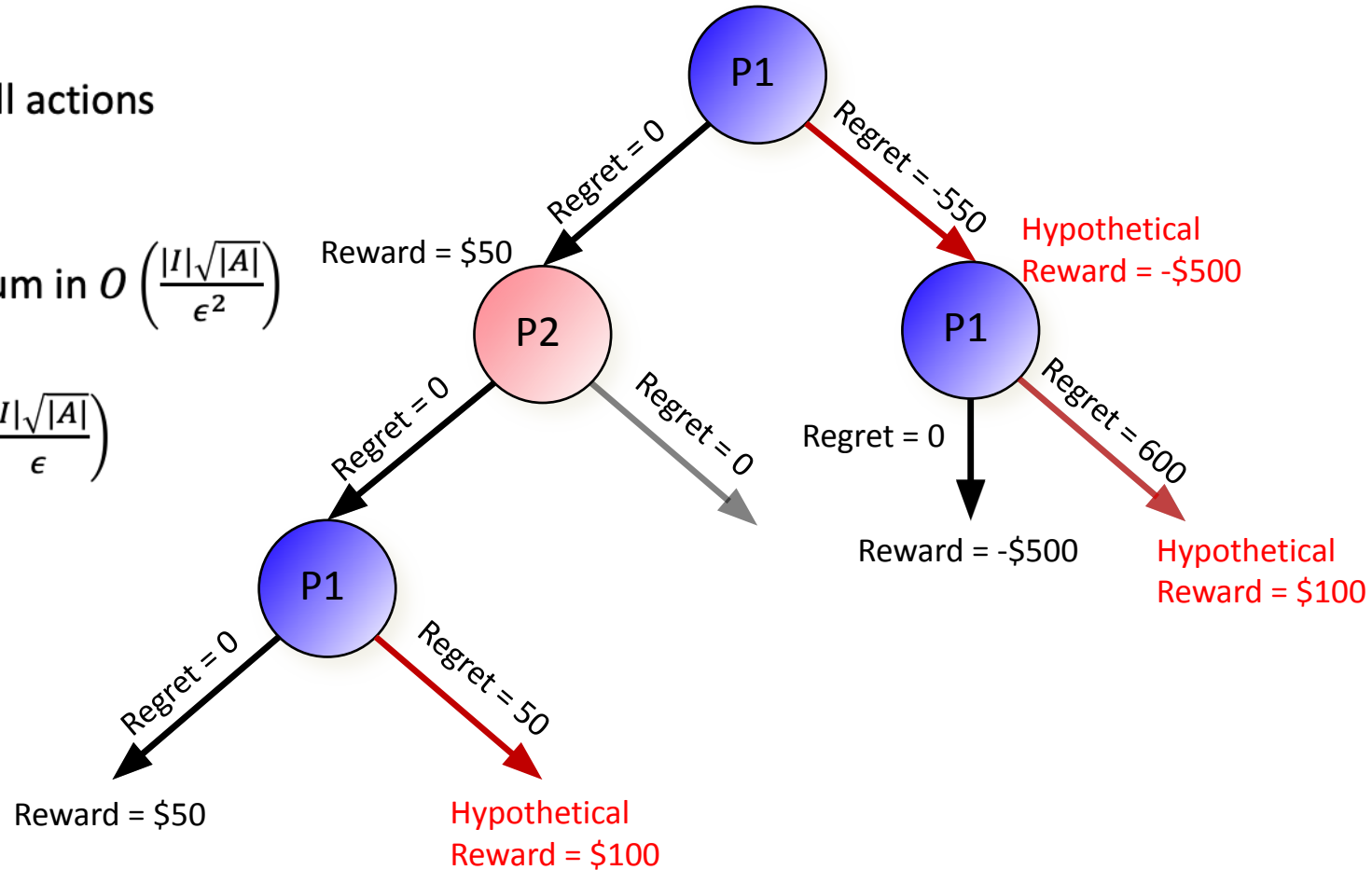
Counterfactual Regret Minimization (CFR)

[Zinkevich *et al.* NeurIPS-07]

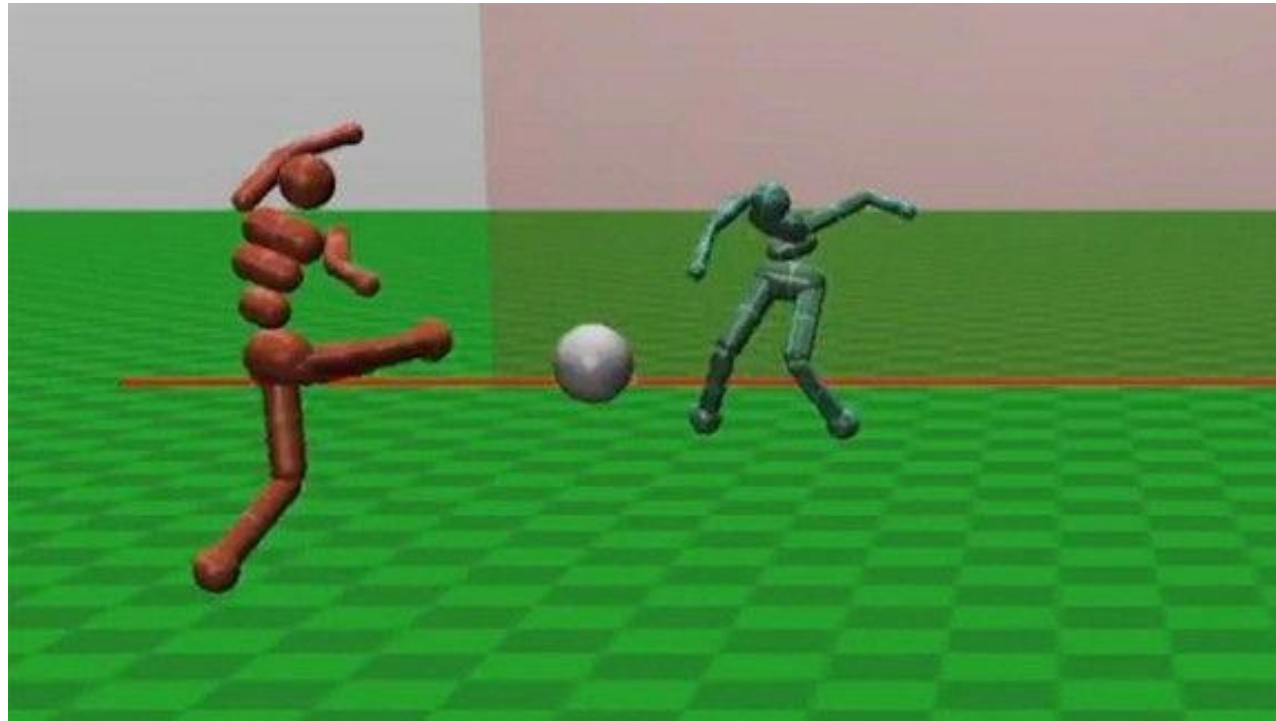
Similar, but takes the EV over all actions rather than sampling

Average converges to equilibrium in $O\left(\frac{|I|\sqrt{|A|}}{\epsilon^2}\right)$

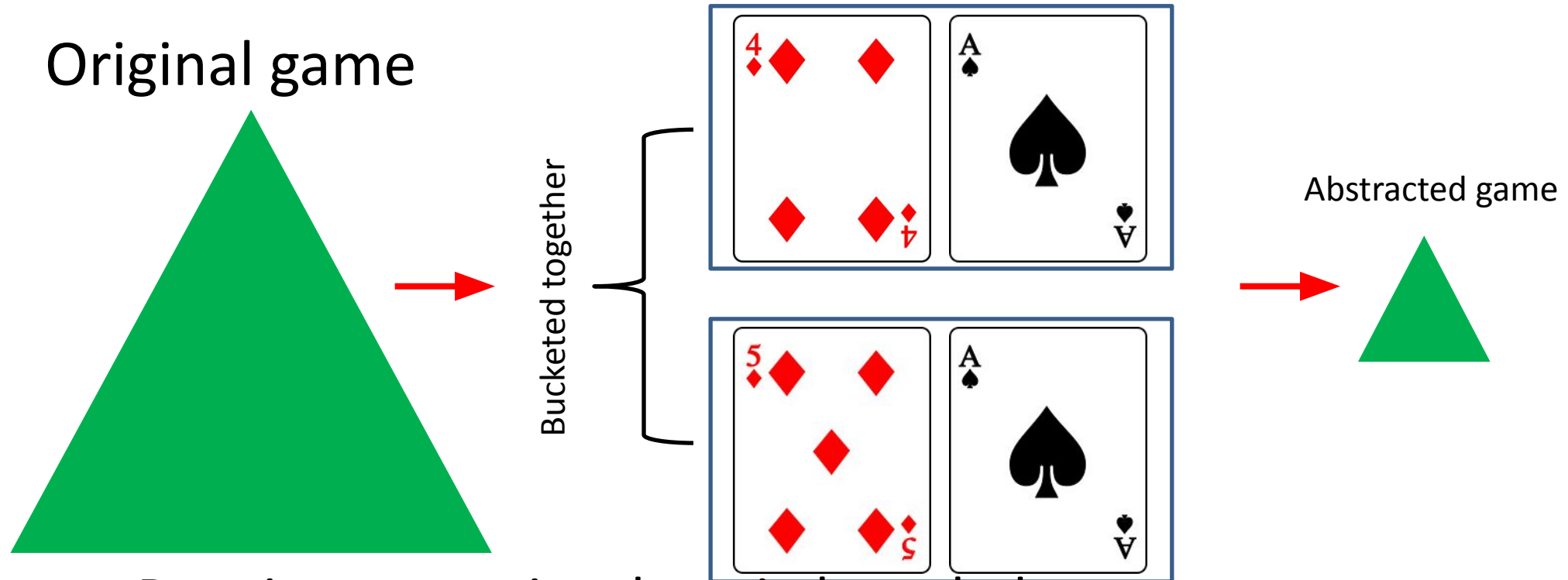
But in practice converge in $O\left(\frac{|I|\sqrt{|A|}}{\epsilon}\right)$



Extending CFR to Large Games



Prior Approach: Abstraction in Games

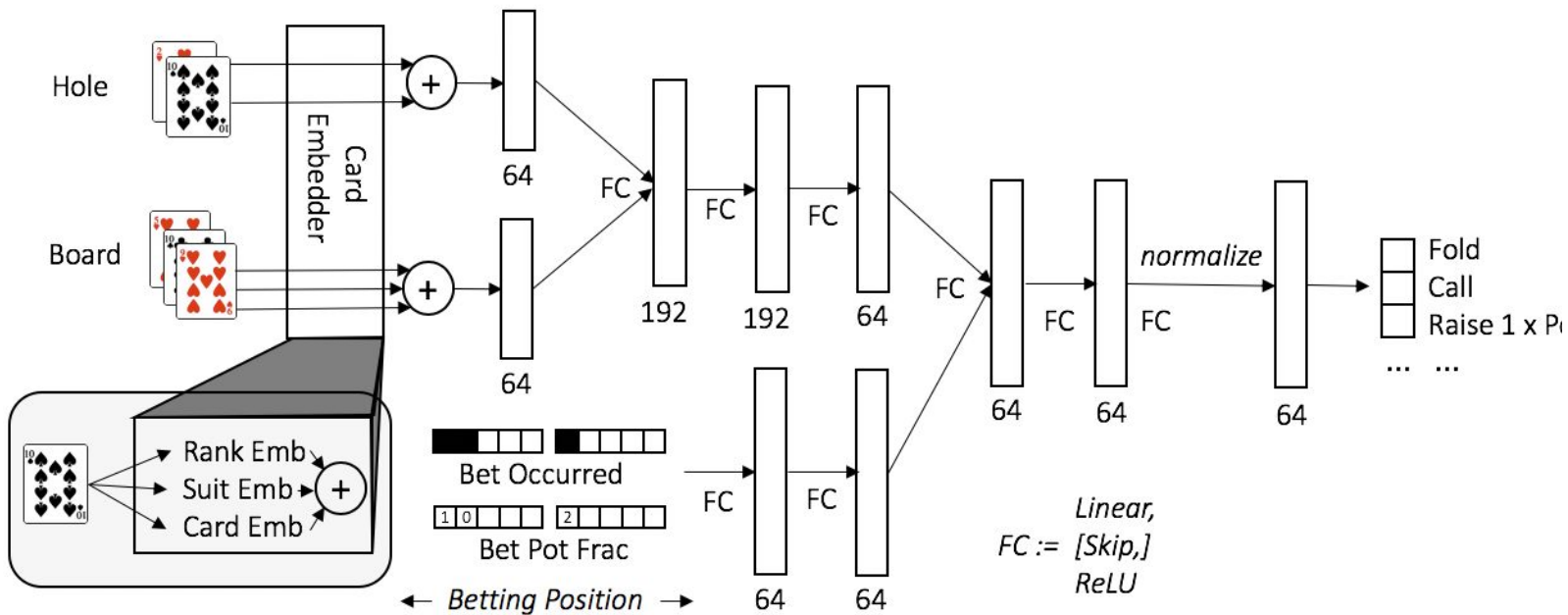


- Requires extensive domain knowledge
 - Several papers written on how to do abstraction just in poker
[\[Johanson et al. AAMAS-13, Ganzfried & Sandholm AAAI-14\]](#)
 - Difficult to extend to other games

Deep CFR / DREAM

[Brown et al. ICML-19; Steinberger, Lerer, Brown arXiv-20]

- Replaces abstraction with neural network approximation of regrets
- Deep CFR / DREAM require far less domain knowledge



Searching for a better strategy in real time



Image Credit: UC Berkeley CS-188 Lecture 6

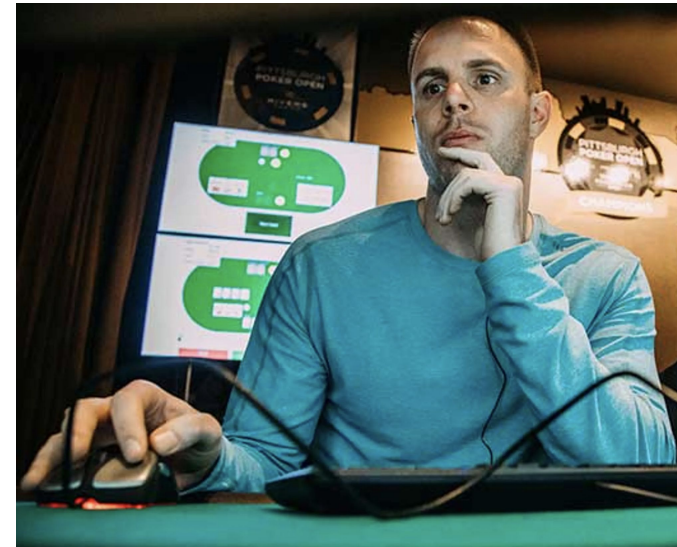
Annual Computer Poker Competition

- Each year, research labs would make poker bots and play them against each other.
- It turned into a competition of scaling models:
 - **2012:** 5,000 buckets
 - **2013:** 30,000 buckets
 - **2014:** 90,000 buckets
 - **2015:** 600,000 buckets

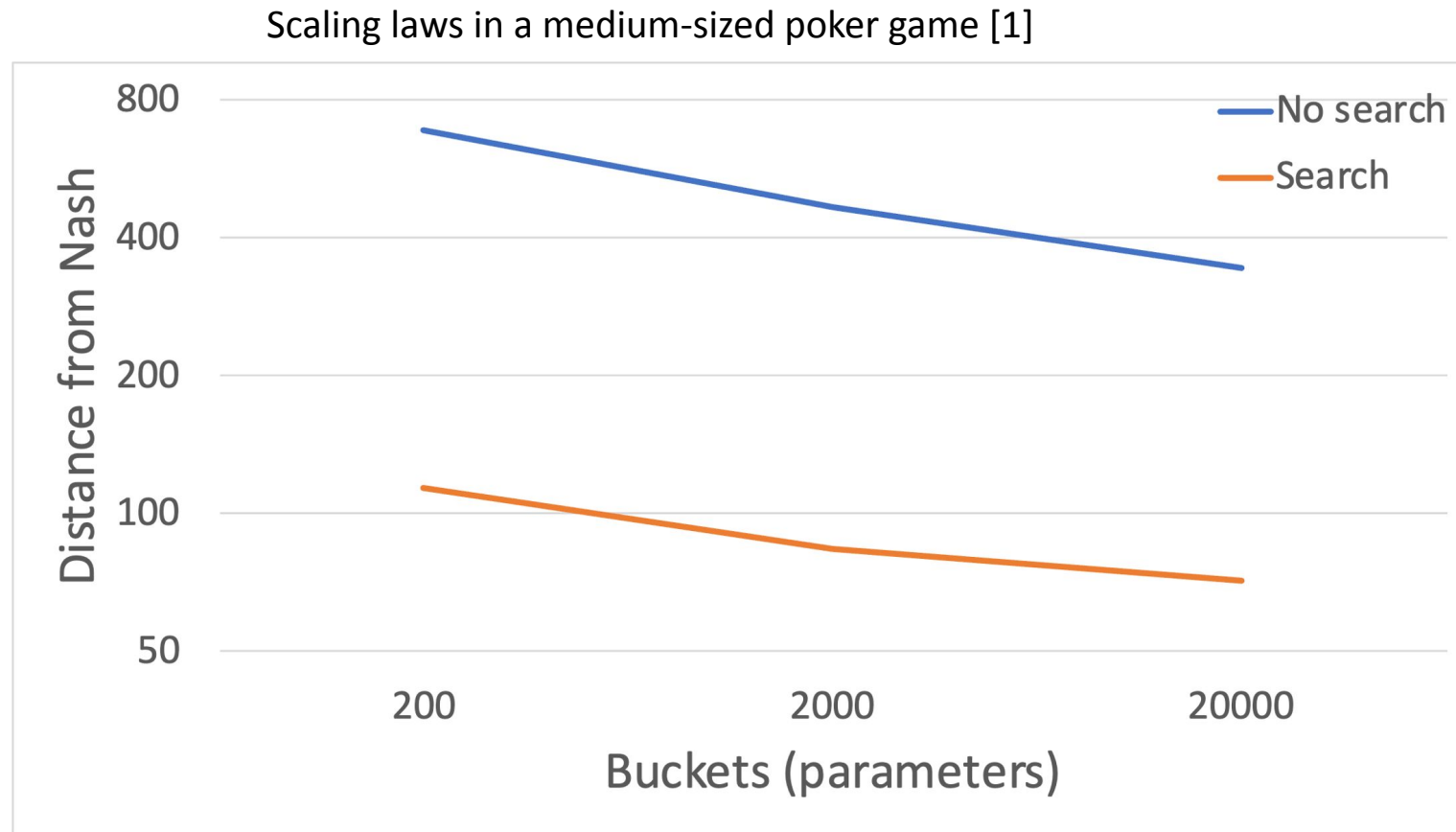


2015 Brains vs. AI Poker Competition

- In 2015 we (CMU) challenged 4 top poker pros to an 80,000-hand poker competition
- \$120,000 in prize money to incentivize them
- Our bot (Claudico) lost by 9.1 bb/100



The importance of search in poker



[1] "Safe and Nested Subgame Solving in Imperfect-Information Games." Brown & Sandholm. NeurIPS 2017 Best Paper.

2017 Brains vs AI Two-Player Poker AI

[Brown & Sandholm Science-17]

- Libratus against 4 top poker pros

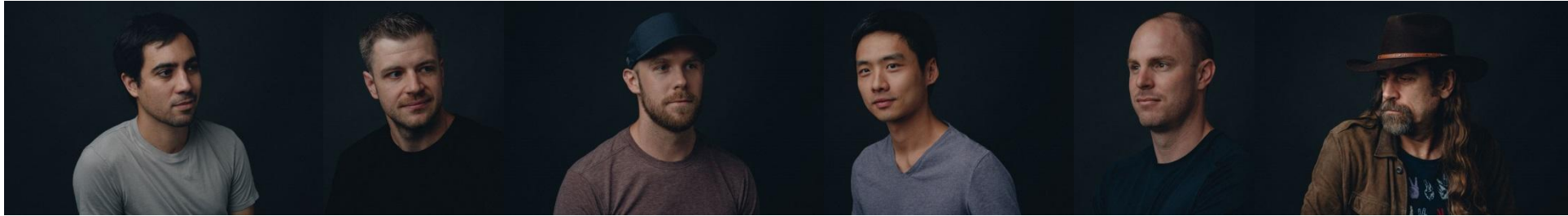


- 120,000 hands of poker
- \$200,000 in prize money
- **Won by 15 bb/100** (Claudico had lost by 9 bb/100)
 - P-value ≈ 0.0002
- Each human lost individually to Libratus

2019 Pluribus Six-Player Poker AI

[Brown & Sandholm Science-19]

- Pluribus against 15 top pros in *six-player* no-limit Texas Hold'em



- 10,000 hands over 12 days in June 2019
 - Used variance-reduction techniques to decrease luck
 - One bot playing with five humans
- Won with >95% statistical significance
- Cost under \$150 to train**, runs on 28 CPU cores (no GPUs)



Why wasn't search considered important in poker before?

- Cultural factors: researchers wanted the solution for the entire game upfront
- Scaling test-time compute makes experiments more expensive
- Incentives:
 - People were always thinking about winning the next Annual Computer Poker Competition (ACPC)
 - The ACPC limited test-time compute to 2 CPU cores
- **Most important:** people underestimated the difference it would make

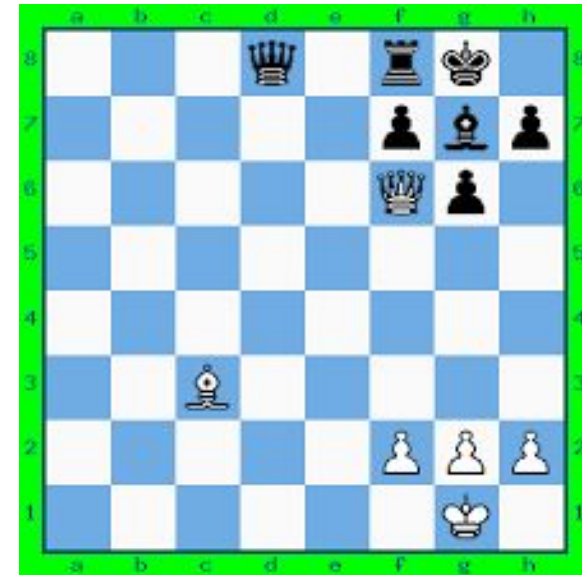
Why is search in imperfect-information games hard?

**Because “states” as traditionally defined
don’t have well-defined values**

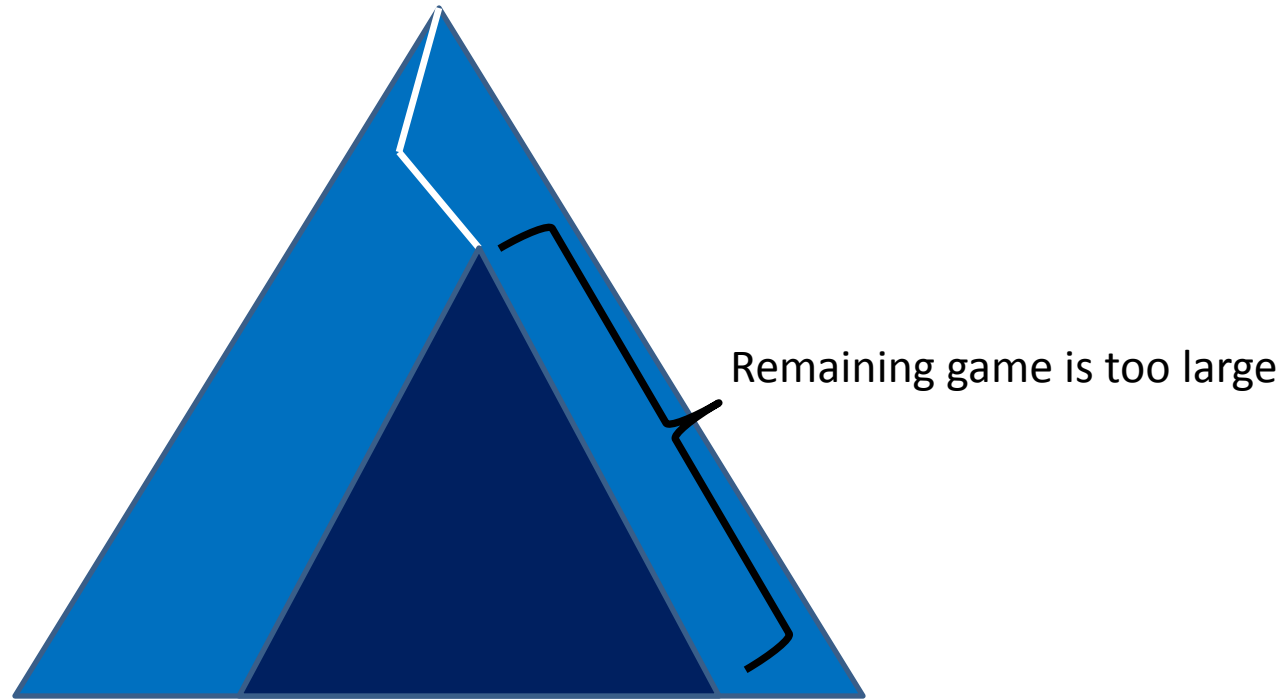
Search in Perfect-Information Games

- In perfect-information games, the **value of a state** is the **unique** value of both players playing optimally from that point forward
- A **value network** takes a state as input and outputs an estimate of the state value

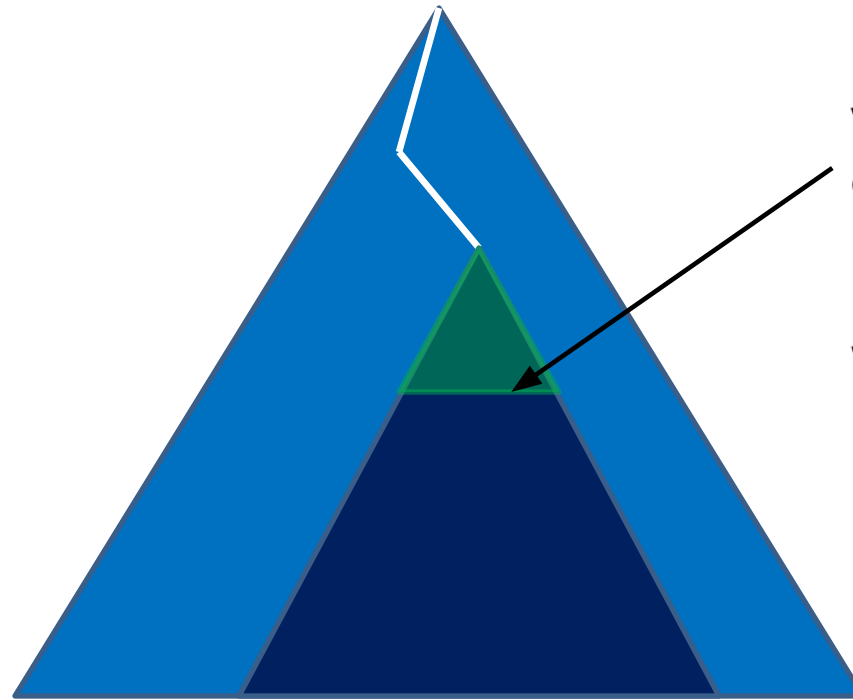
$$f_{white}(\text{state}) = 1$$



Search in Perfect-Information Games



Search in Perfect-Information Games

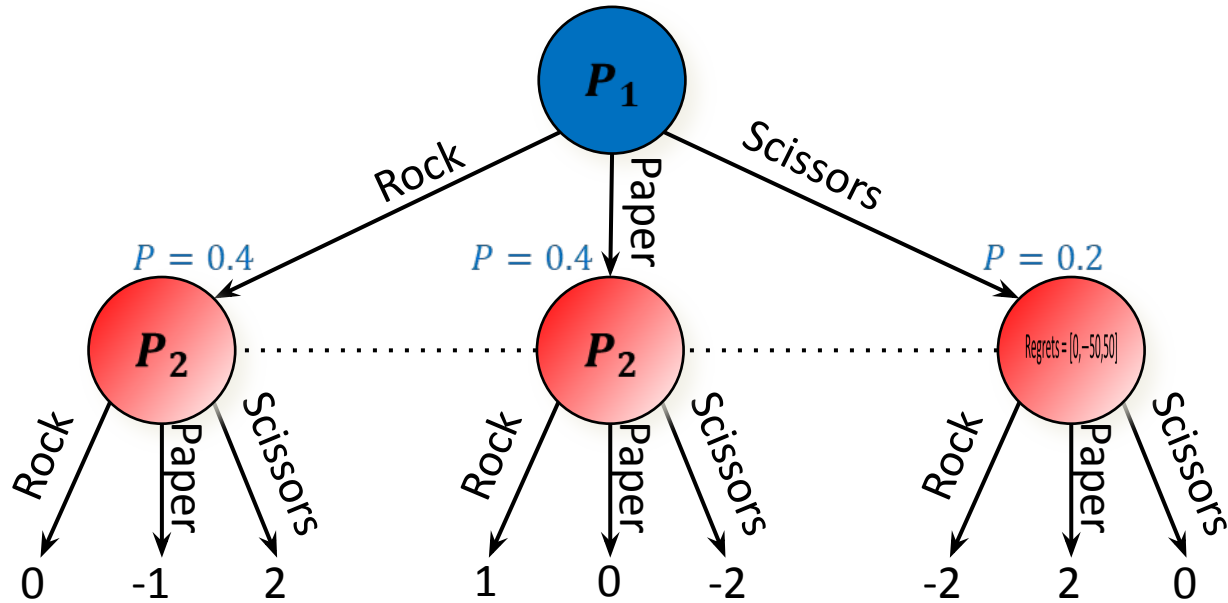


Value substituted at leaf node s is estimate of equilibrium value $\hat{v}(s)$

If estimate is perfect, depth-limited search will find optimal strategy

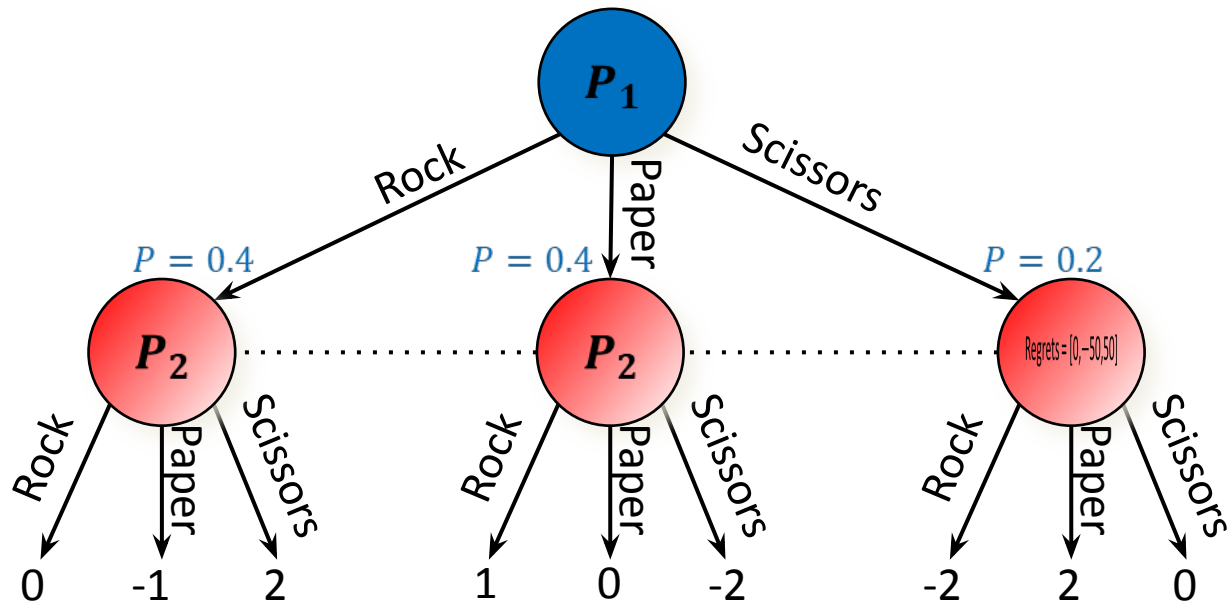
Depth-Limited Search

Rock-Paper-Scissors+

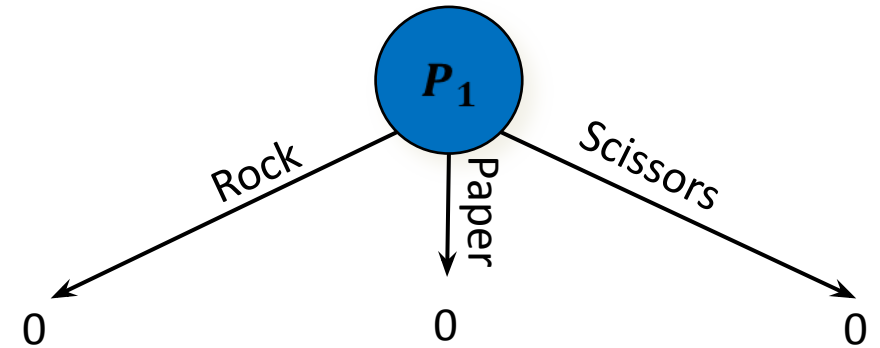


Depth-Limited Search

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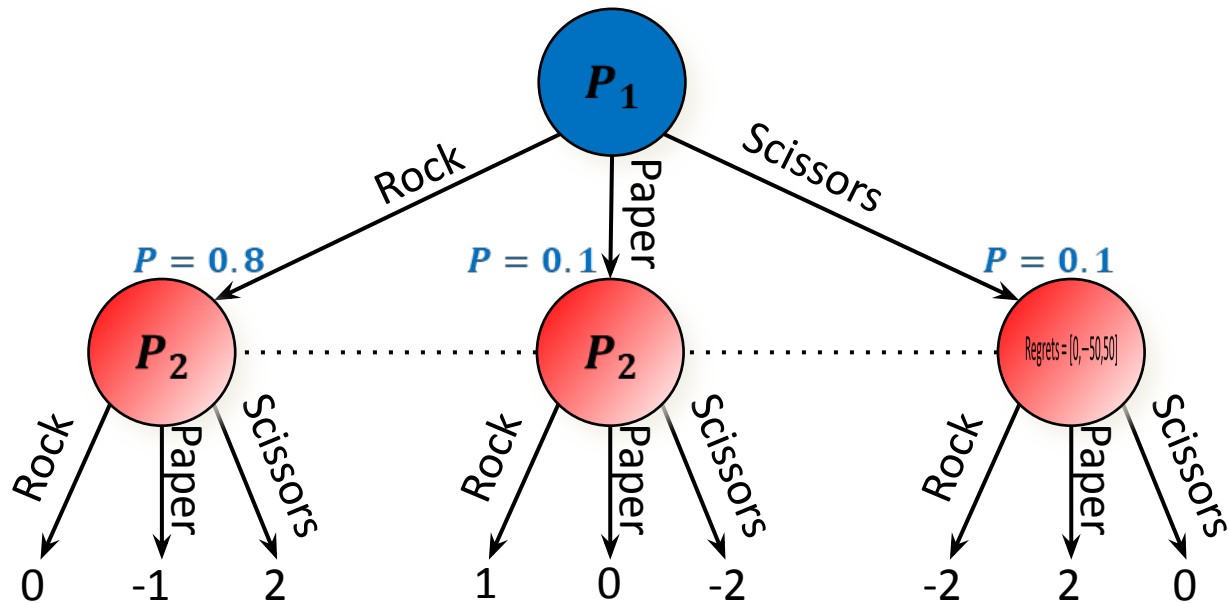


Depth-Limited Rock-Paper-Scissors+

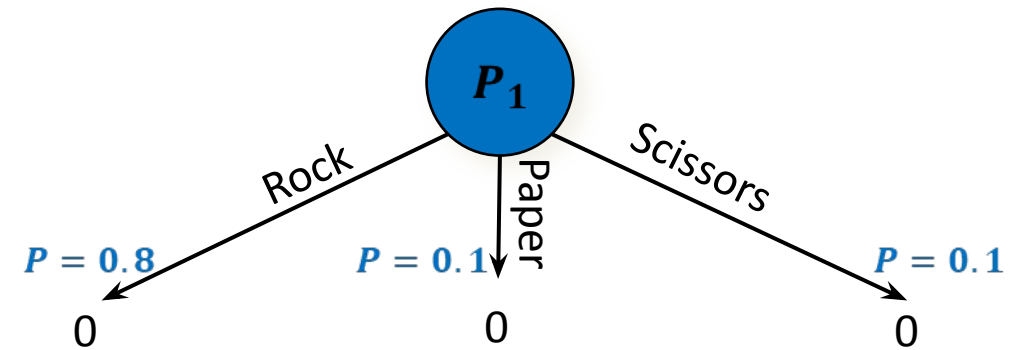


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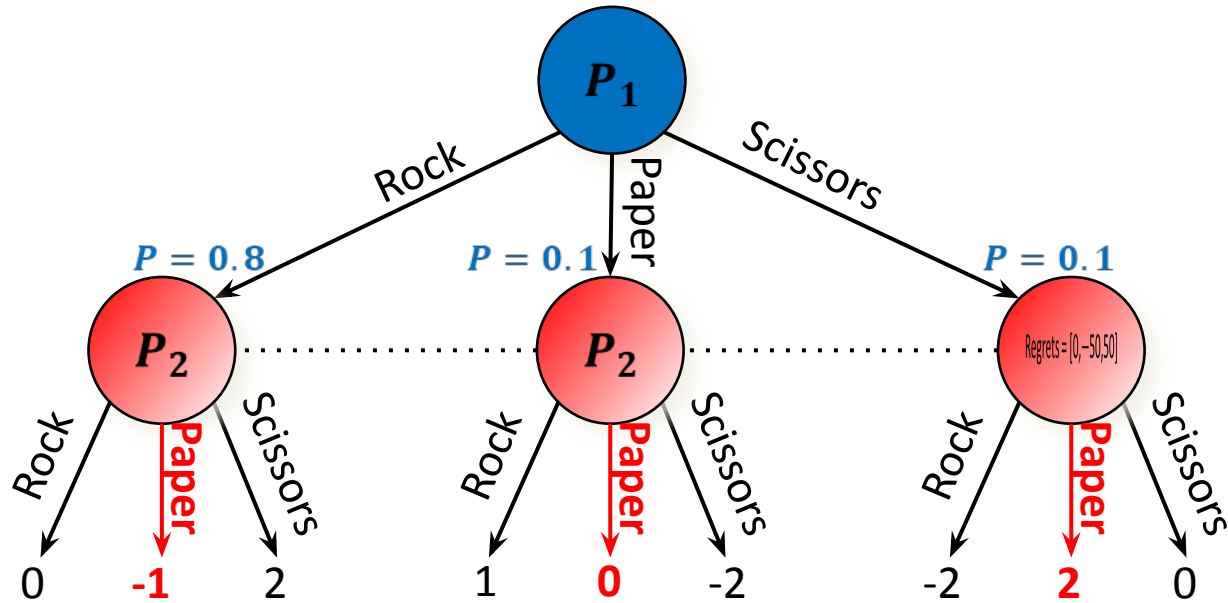


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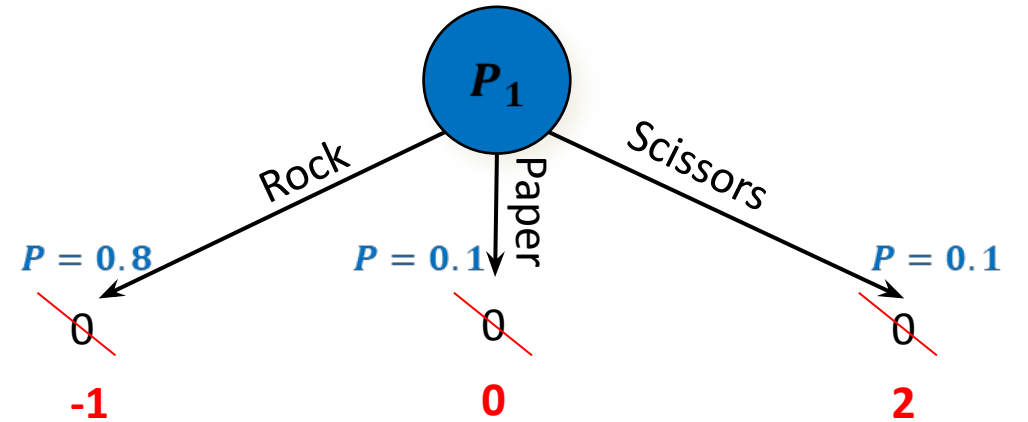


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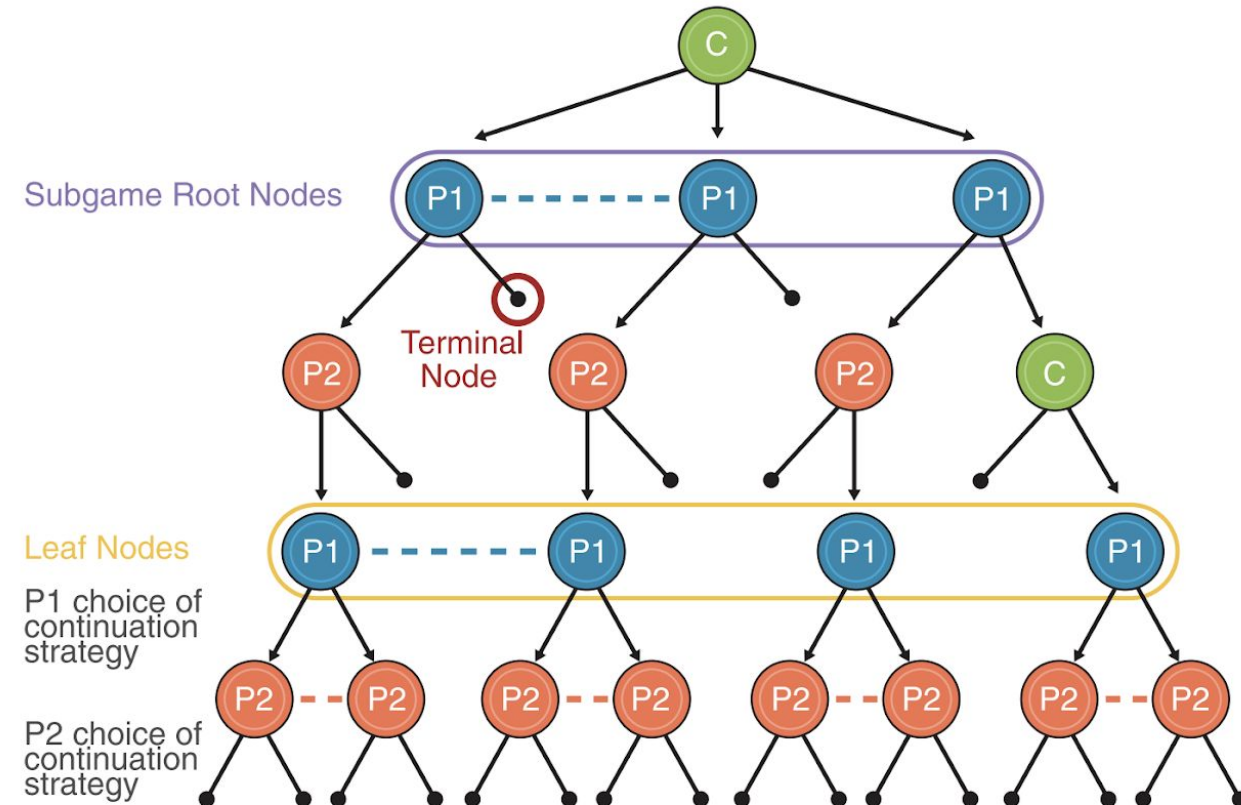
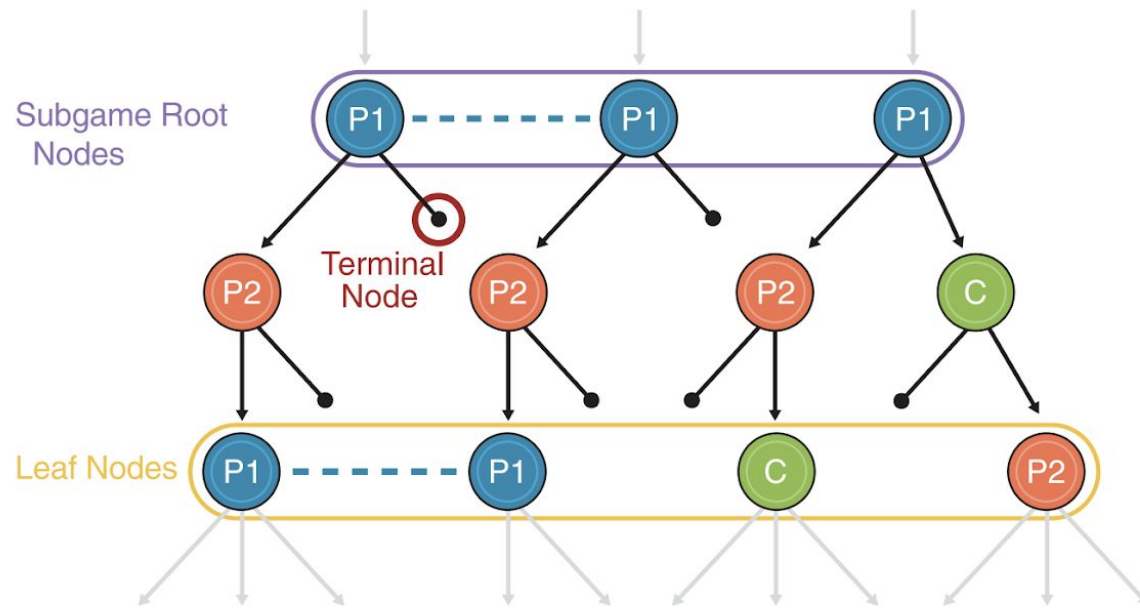


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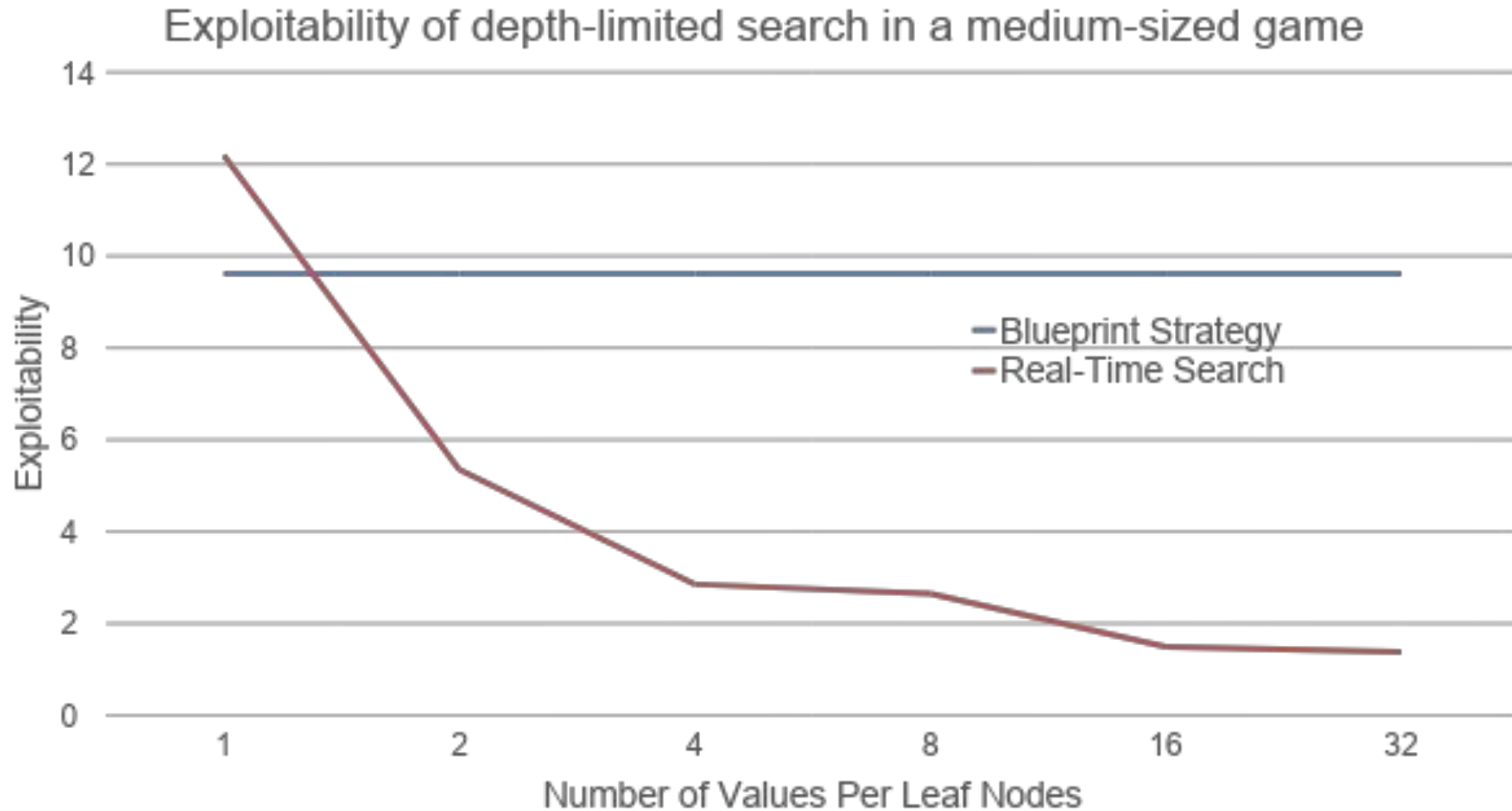


Depth-Limited Search in Pluribus

[Brown, Sandholm, Amos NeurIPS-18; Brown & Sandholm Science-19]

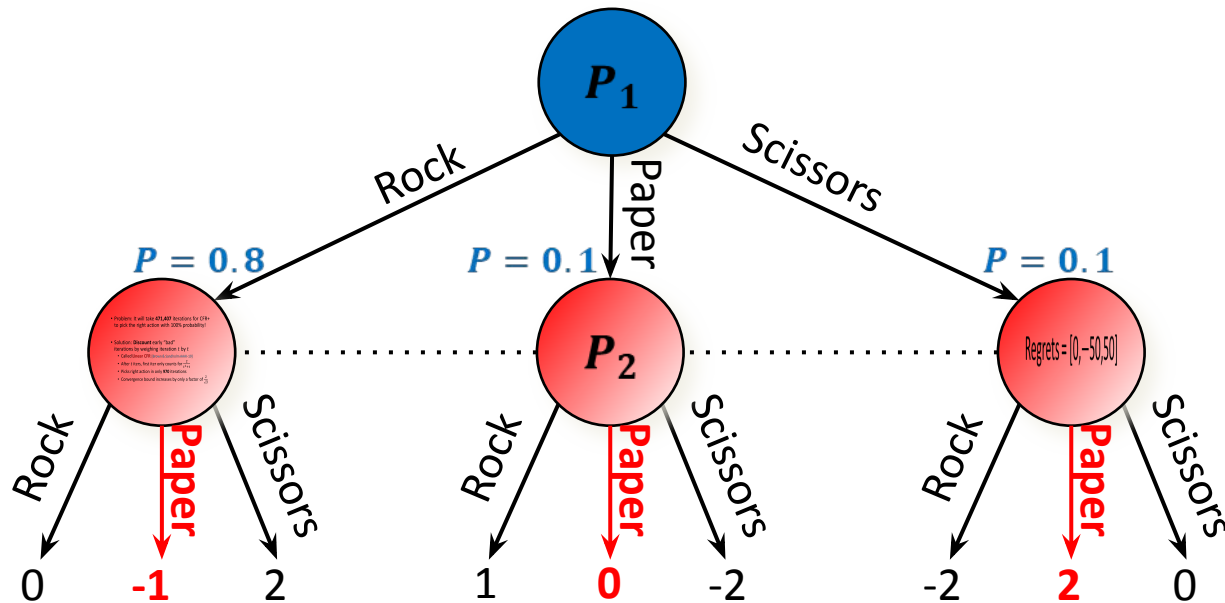


Exploitability Measurements

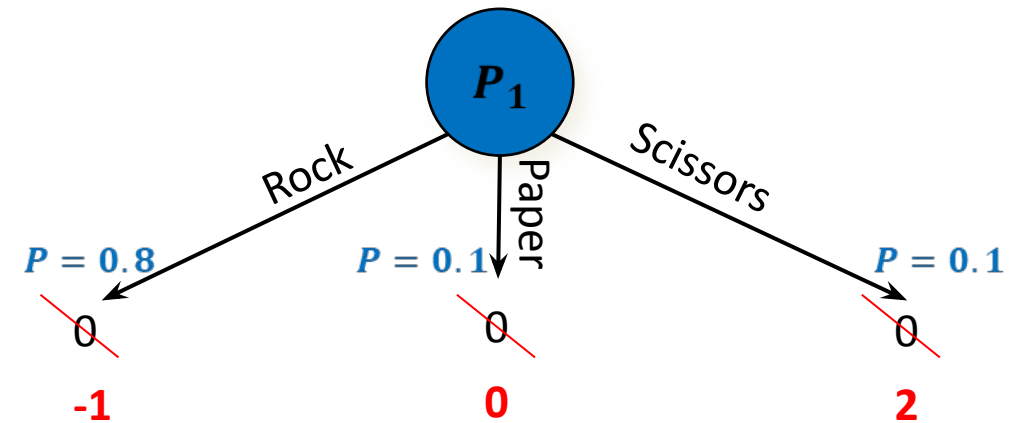


Search in Imperfect-Information Games

Rock-Paper-Scissors+



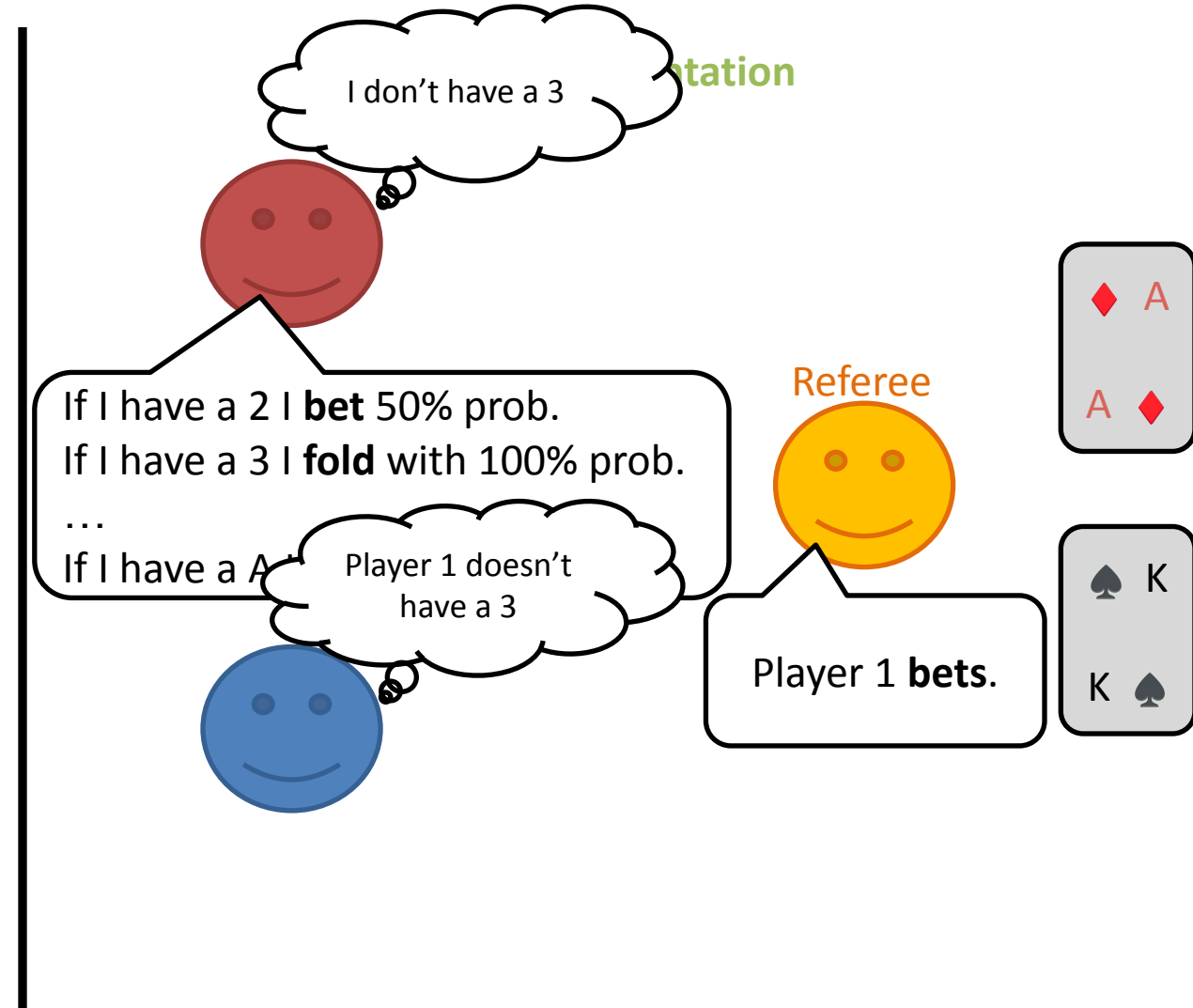
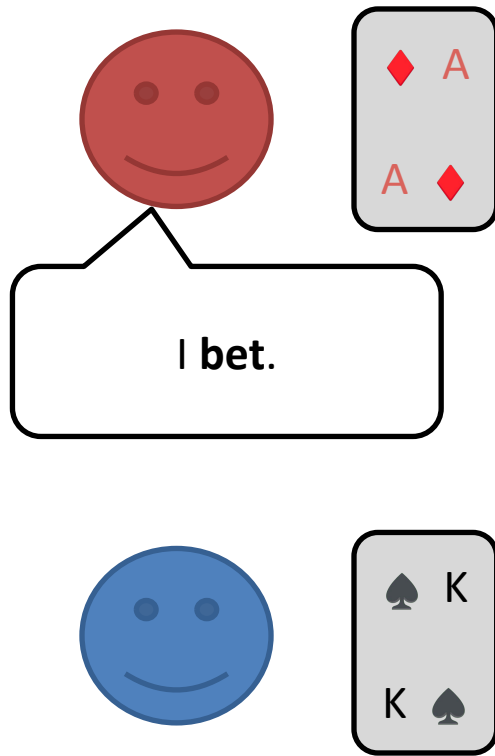
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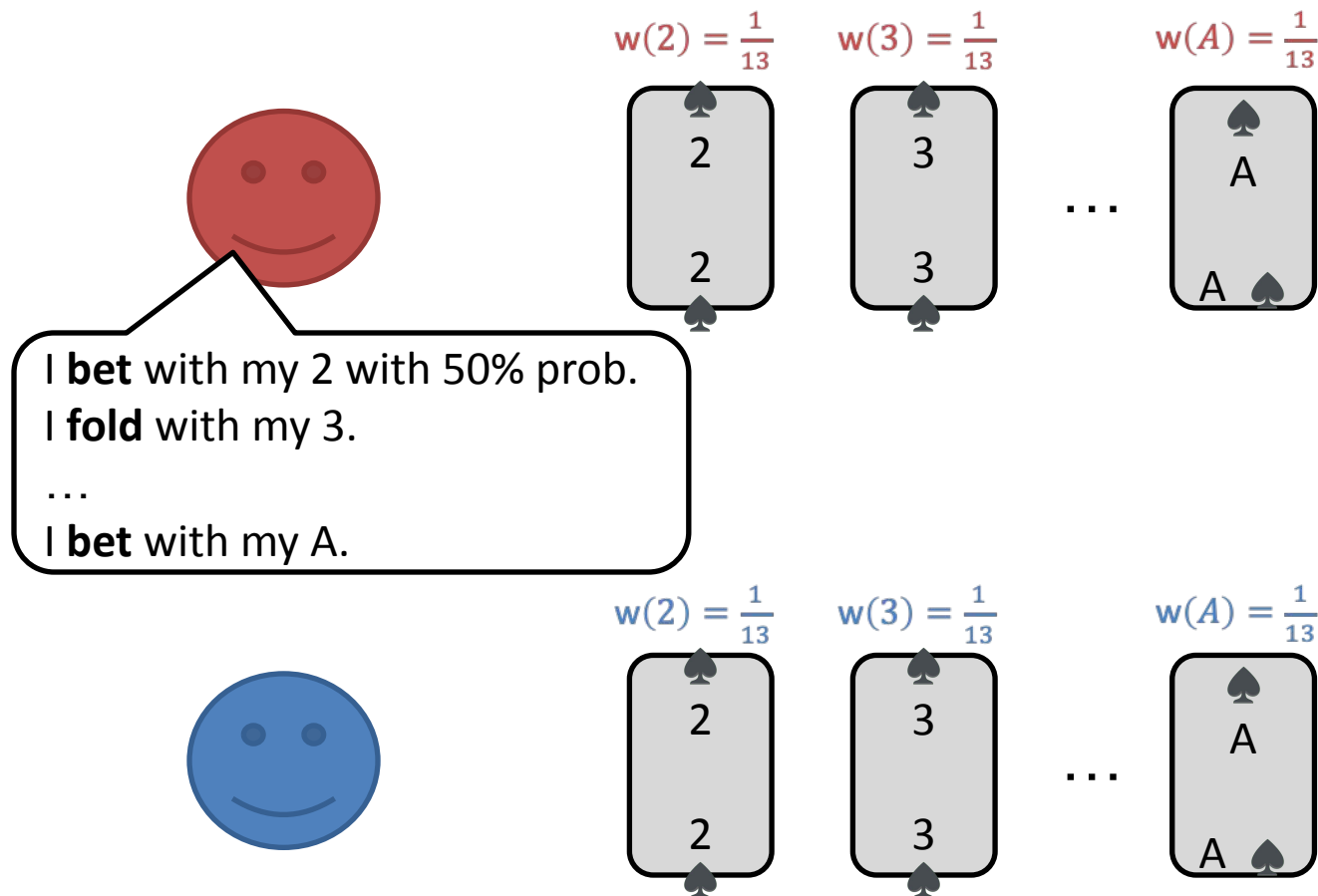
- Another solution: define an imperfect-information game “state” as a **probability distribution over infostates** [Nayyar et al. IEEE-13]
 - $v(\text{Rock})$ is not well-defined
 - $v([0.8 \text{ Rock}, 0.1 \text{ Paper}, 0.1 \text{ Scissors}]) = -0.6$
 - In more complex games, need to include probability distribution for **both** players

Converting imperfect-info games to continuous-state perfect-info games

Discrete Representation



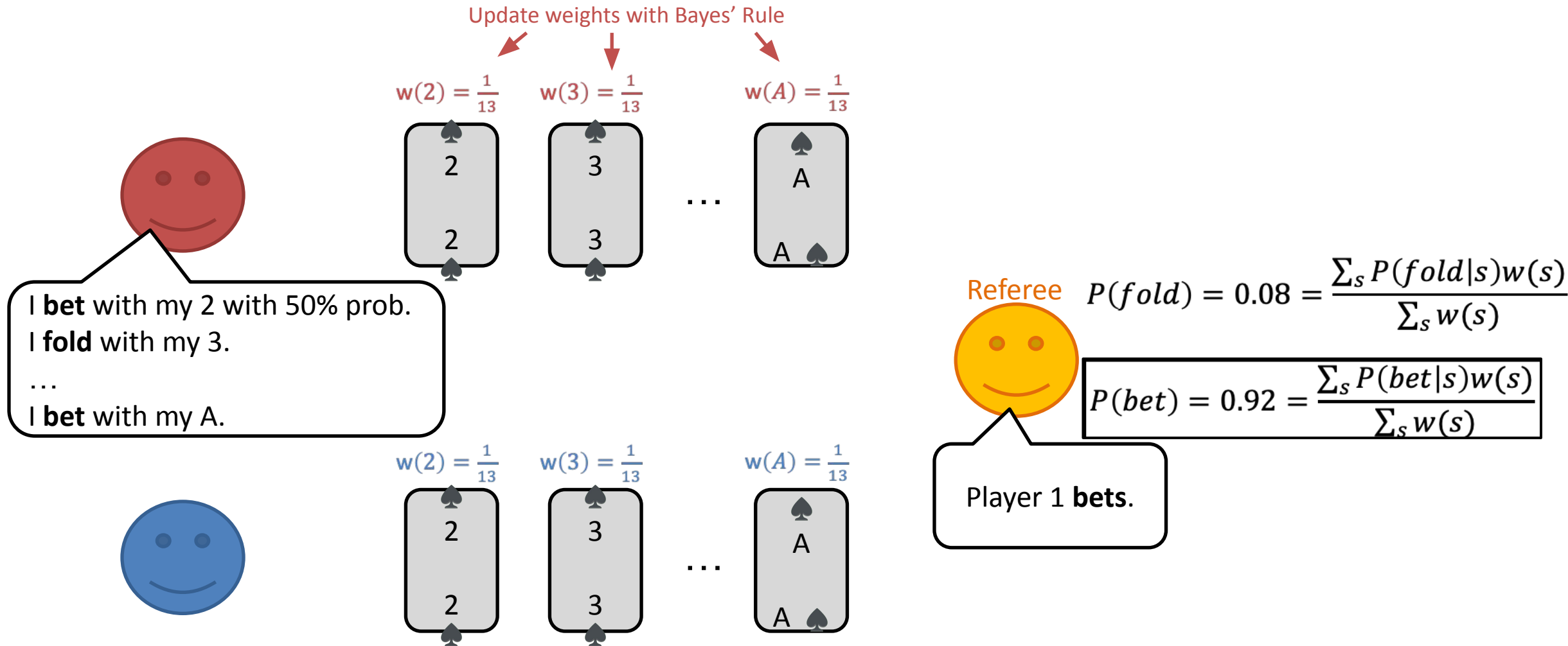
Converting imperfect-info games to continuous-state perfect-info games



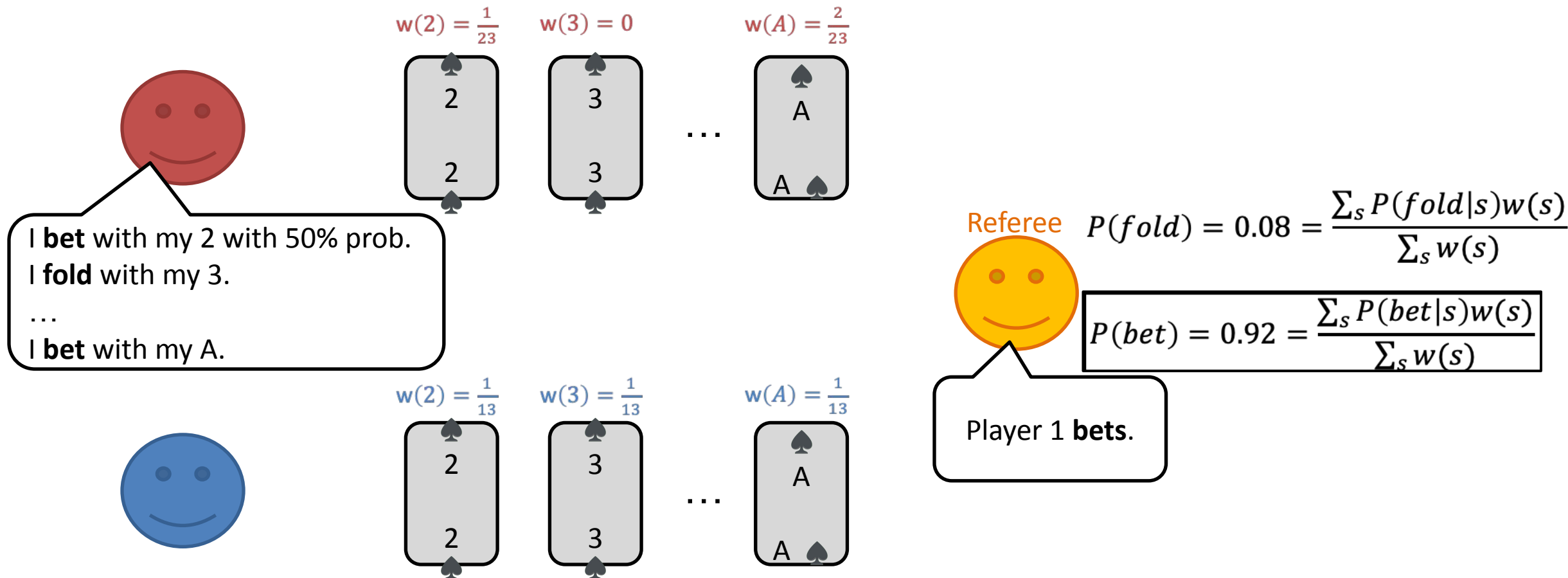
Referee

$$P(\text{fold}) = 0.08 = \frac{\sum_s P(\text{fold}|s)w(s)}{\sum_s w(s)}$$
$$P(\text{bet}) = 0.92 = \frac{\sum_s P(\text{bet}|s)w(s)}{\sum_s w(s)}$$

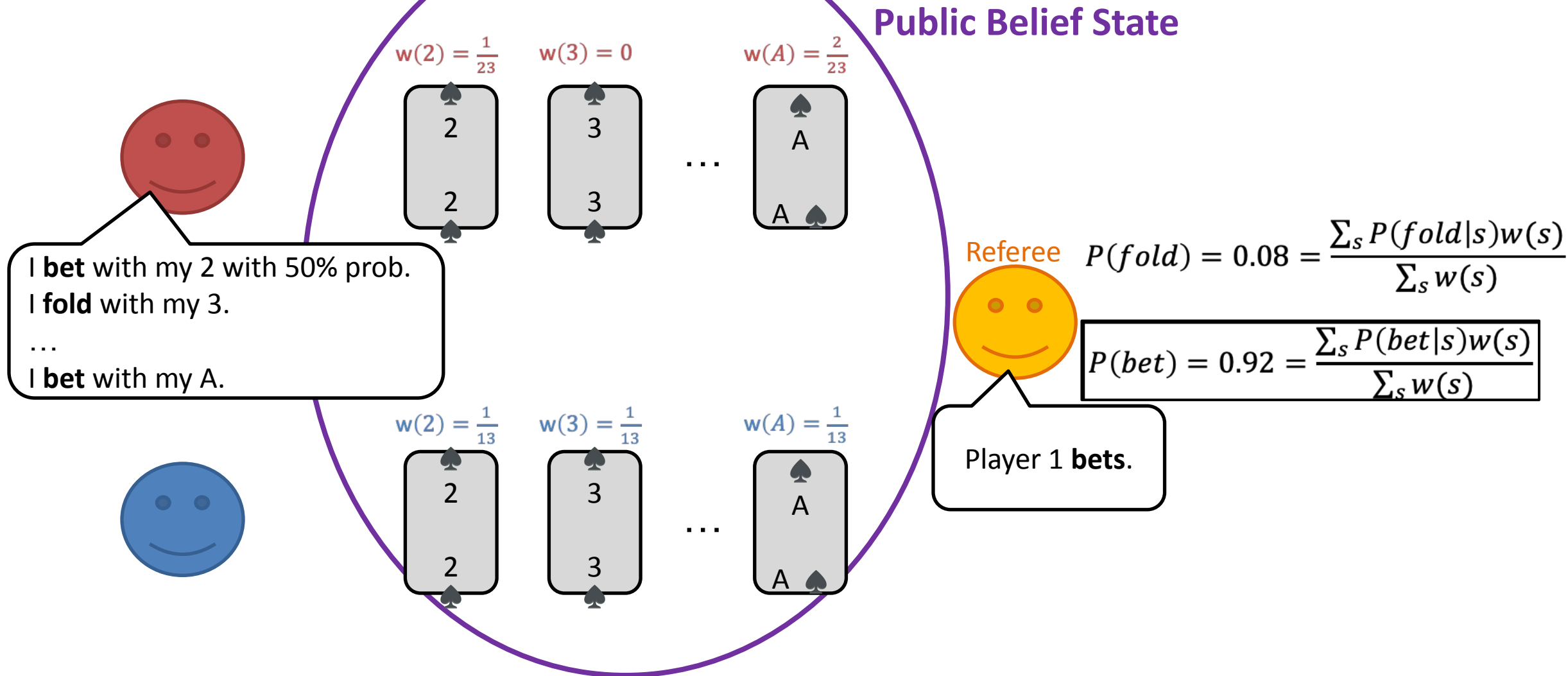
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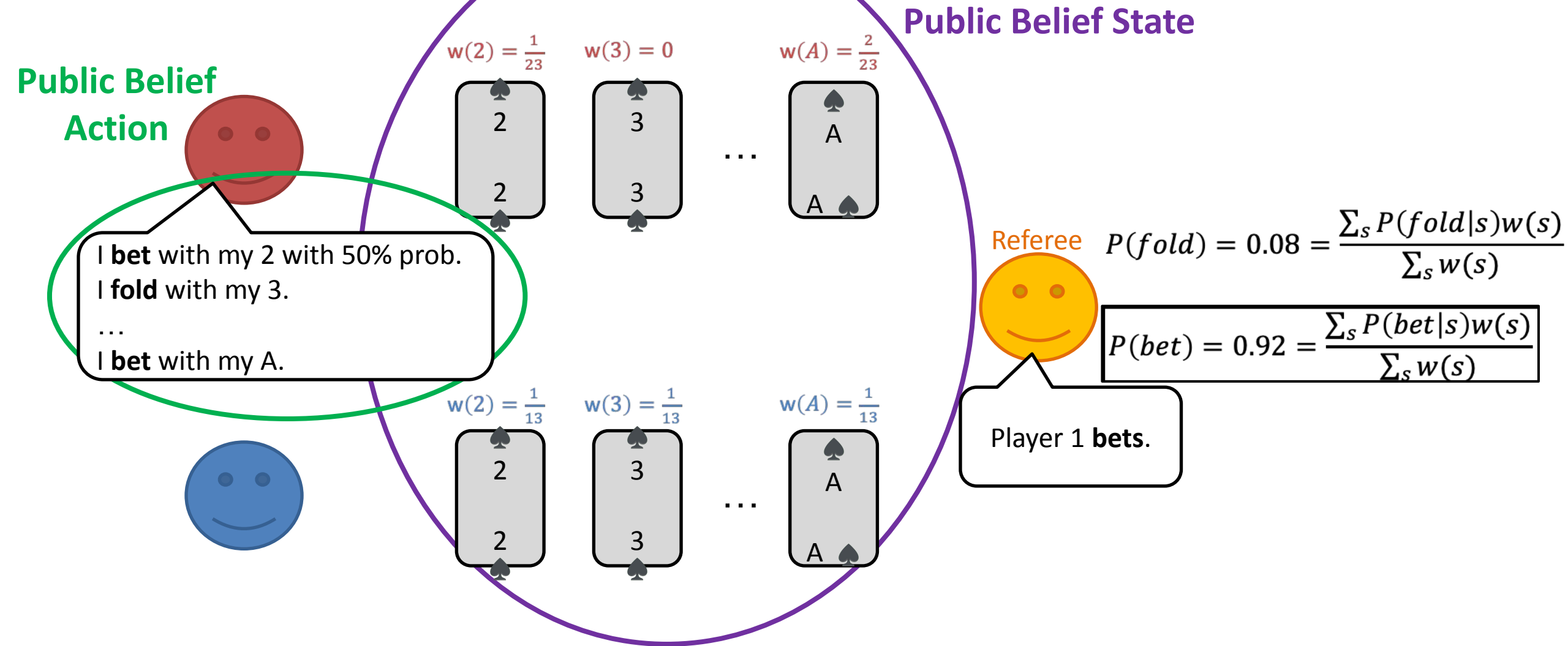
Converting imperfect-info games to continuous-state perfect-info games



Converting imperfect-info games to continuous-state perfect-info games



Converting imperfect-info games to continuous-state perfect-info games



Search in ReBeL

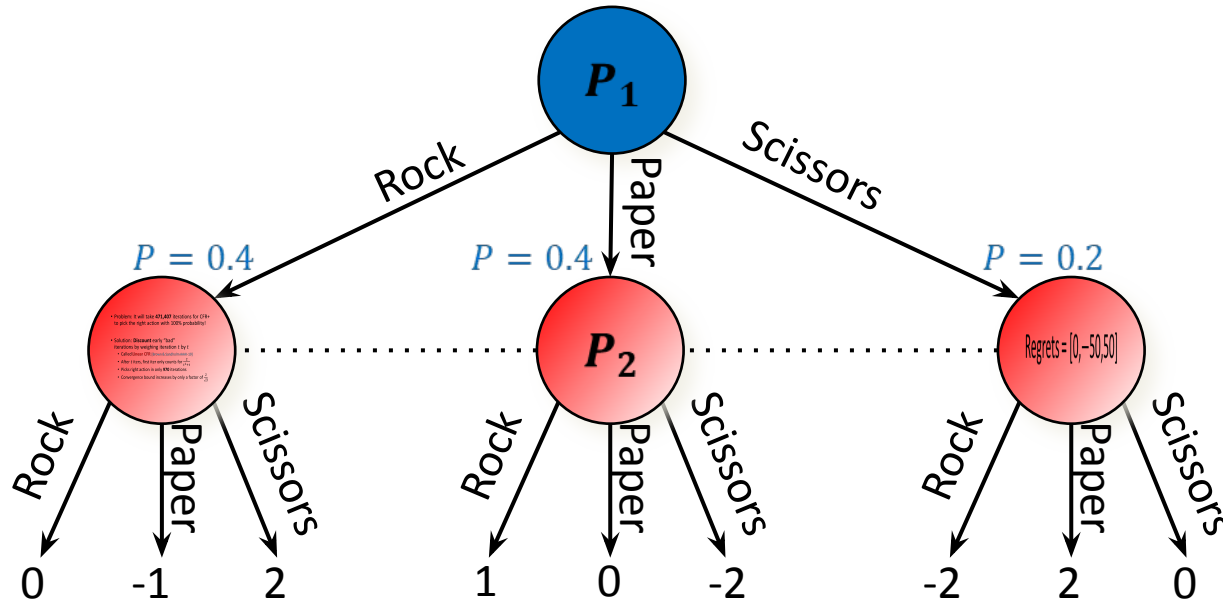
- We've shown all imperfect-information games can be converted into perfect-information games! Can we now run AlphaZero?
- In theory, yes. In practice, no.
 - Action space is continuous with potentially *thousands* of dimensions
 - AlphaZero's Monte Carlo tree search would be completely intractable

Search in ReBeL

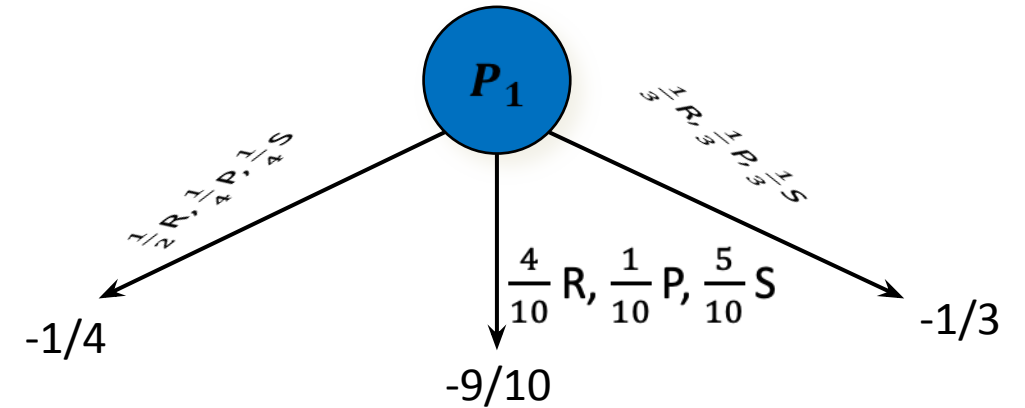
- But! The continuous action space has special structure
 - Known as a “bilinear saddle point problem”
- Due to convexity, we can solve subgames via gradient descent
 - Regret minimization is similar to gradient descent

Search in Belief Representation with MCTS

Rock-Paper-Scissors+

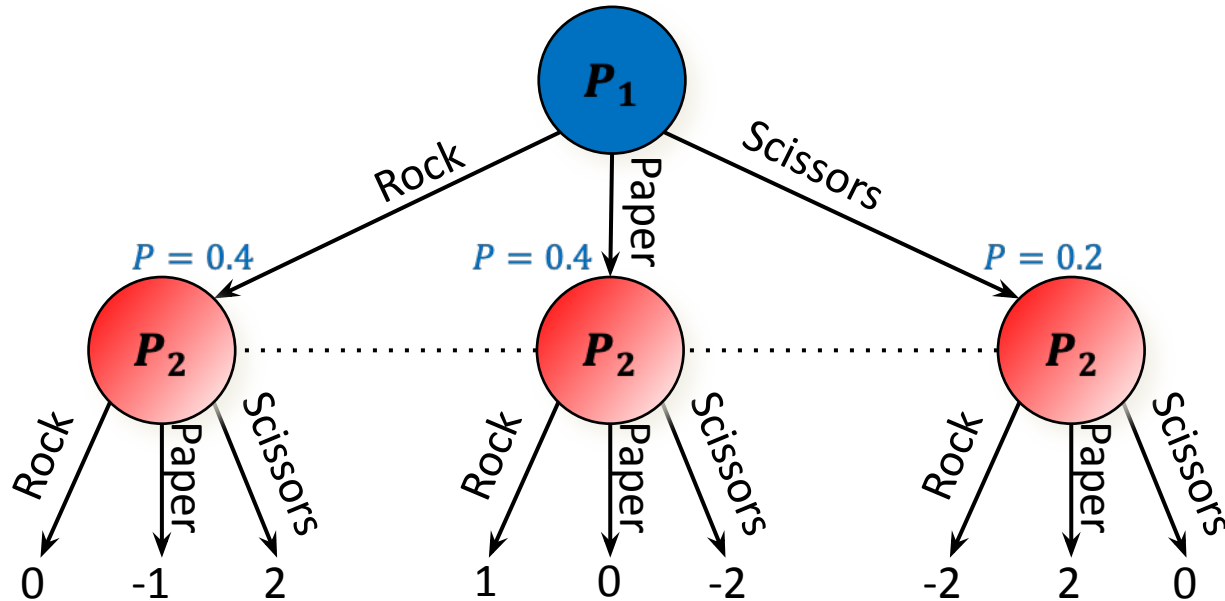


Search in Rock-Paper-Scissors+

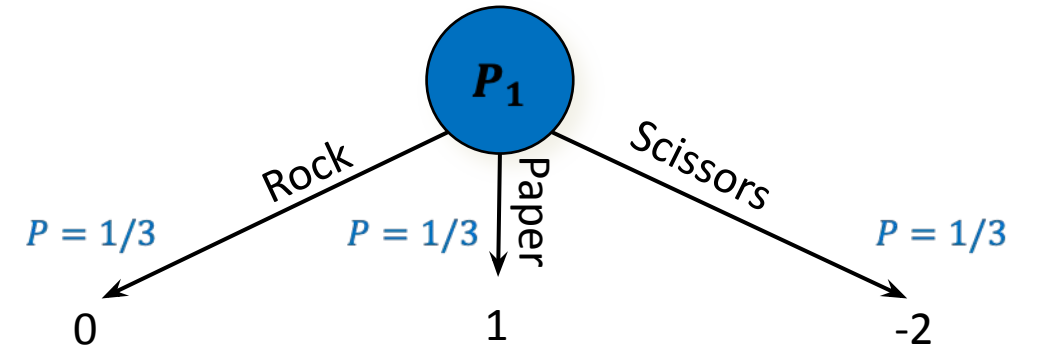


Search in Discrete Representation with Regret Minimization

Rock-Paper-Scissors+

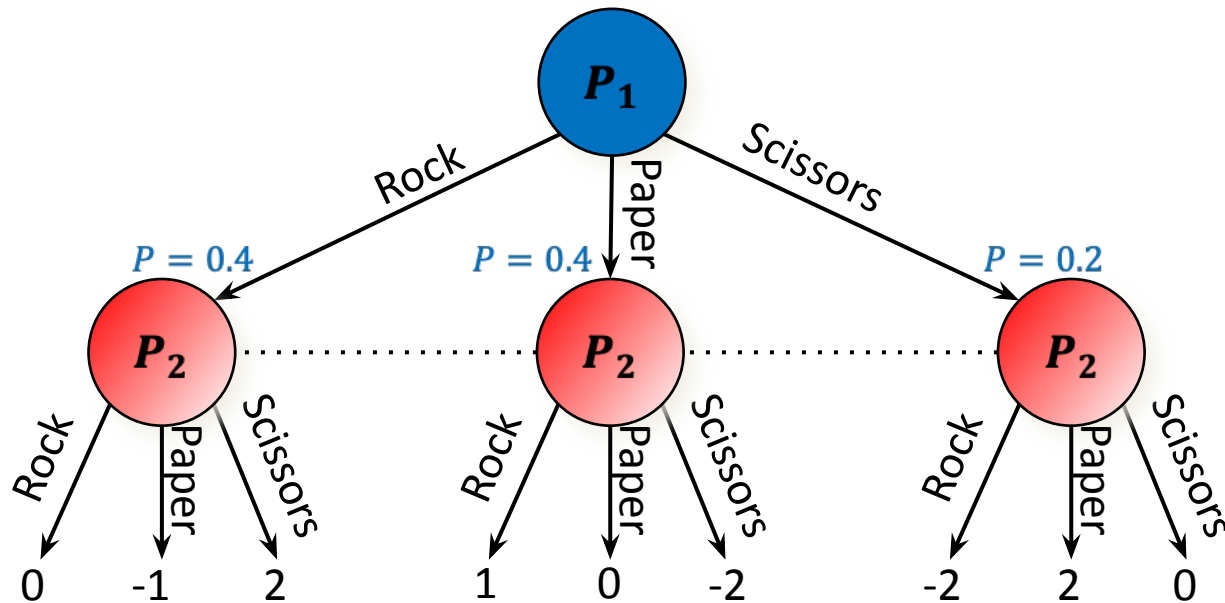


Search in Rock-Paper-Scissors+

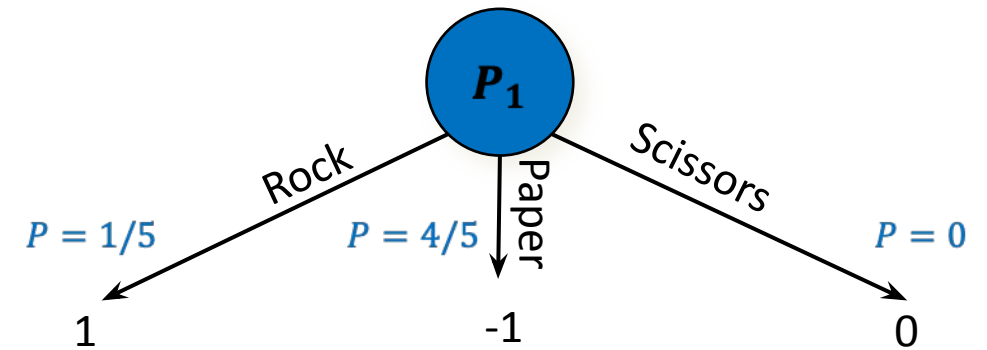


Search in Discrete Representation with Regret Minimization

Rock-Paper-Scissors+



Search in Rock-Paper-Scissors+



- Theorem: the **average policy over all iterations** converges to a **Nash equilibrium**
[Hart & Mas-Colell '00]
 - Equivalently, playing a **random iteration's** policy results in a **Nash equilibrium in expectation**

Is there a general way to scale
inference compute in LLMs?

Scaling Scaling Laws with Board Games (Hex)

[Andy L. Jones, arXiv-2021]

Knowing now that compute can be spent in two places, at train time and test time, the immediate question is: how do these two budgets trade off? This is illustrated in Fig. 9, which shows that the trade-off is linear in log-compute: **for each additional 10× of train-time compute, about 15× of test-time compute can be eliminated**, down to a floor of a single-node tree search.

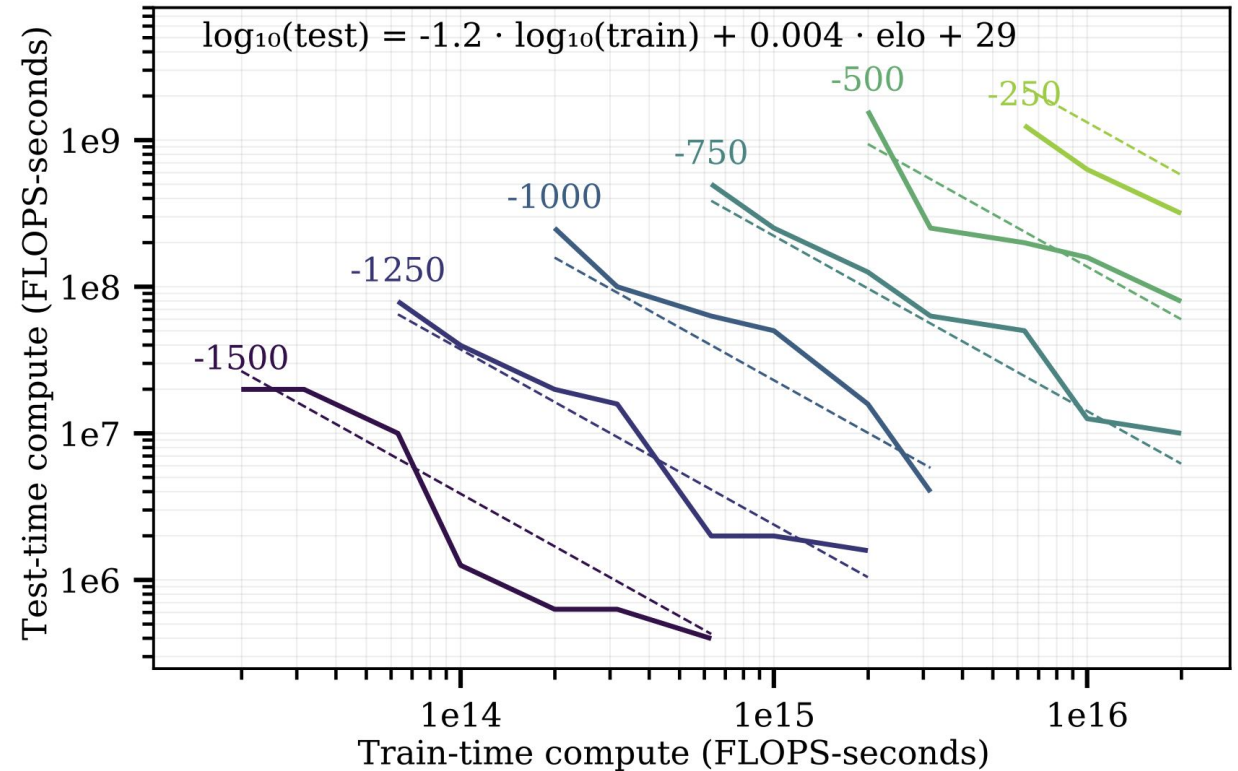
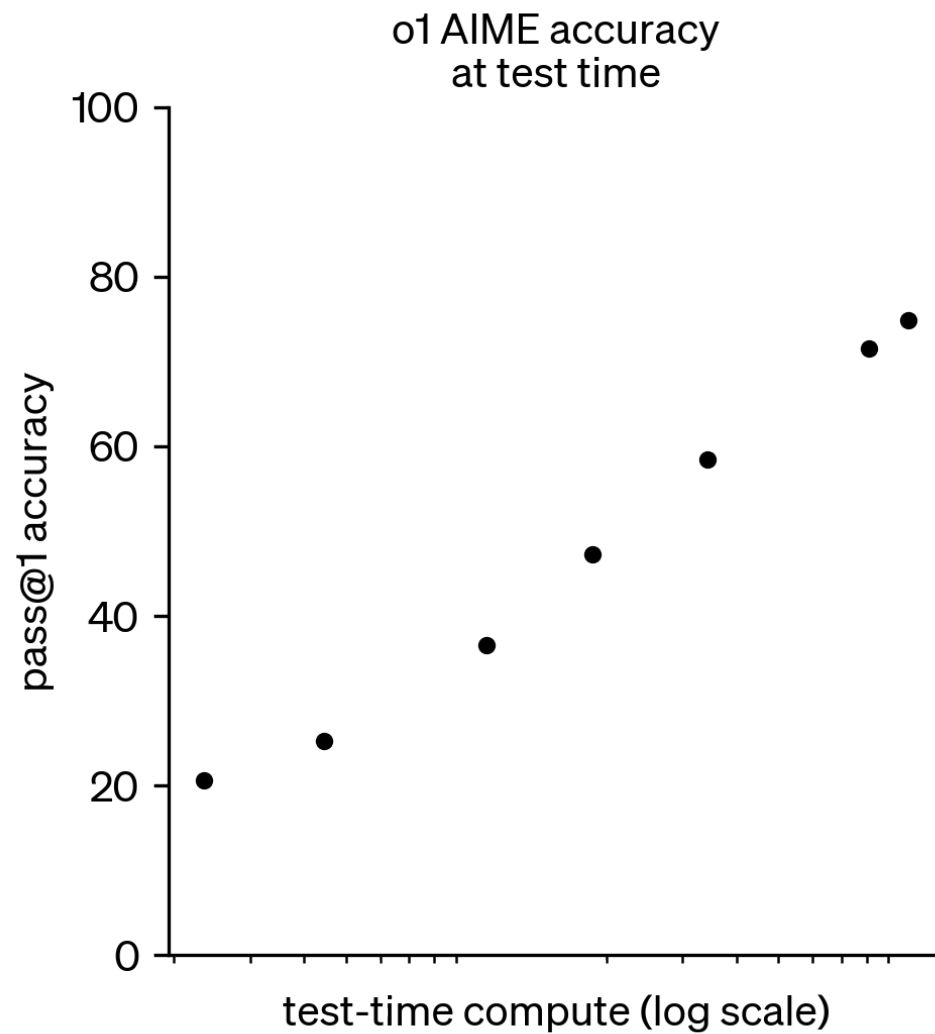
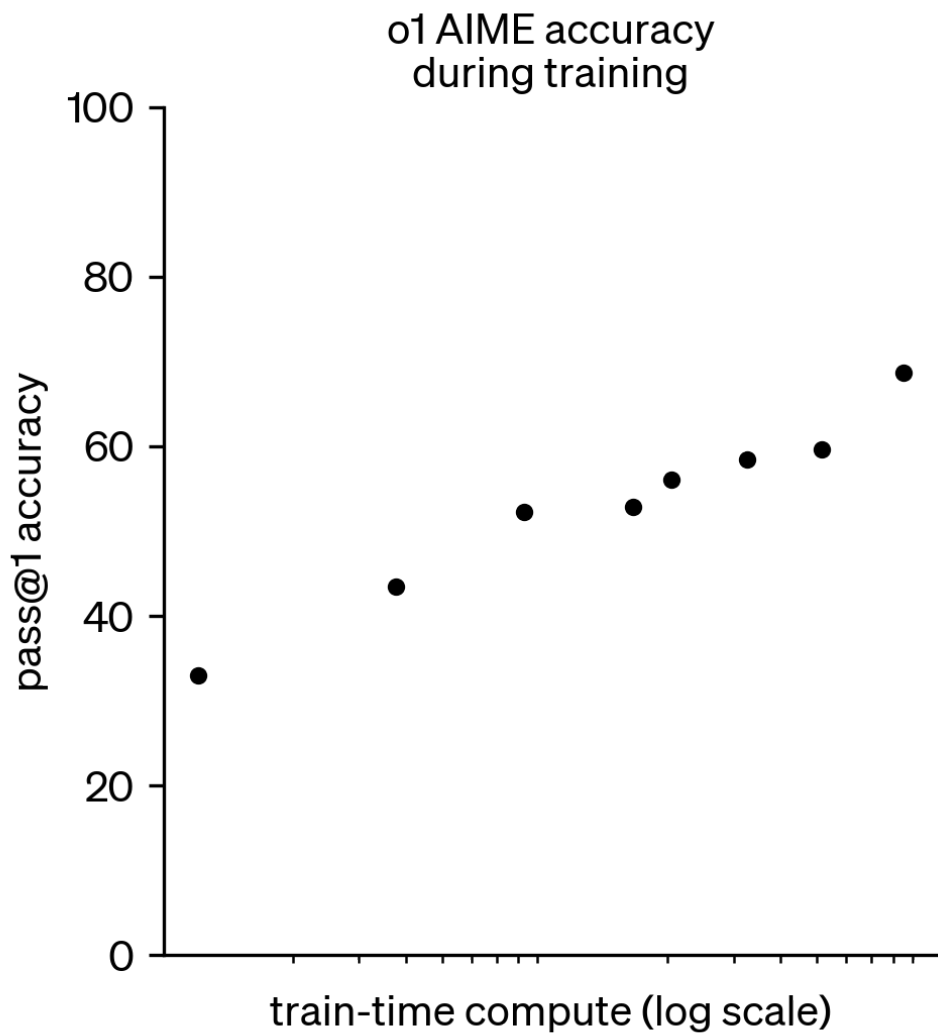
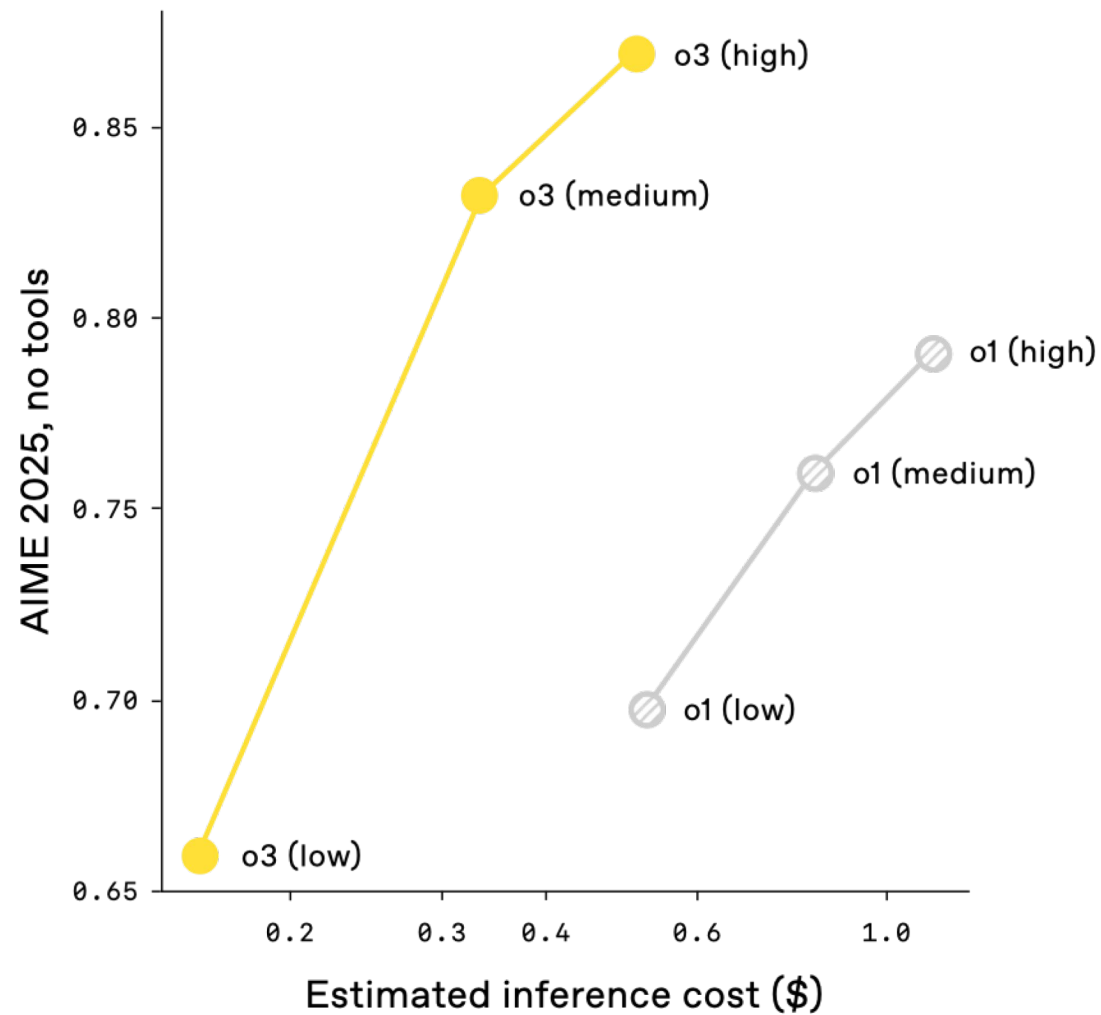


Fig. 9. The trade-off between train-time compute and test-time compute. Each dotted line gives the minimum train-test compute required for a certain Elo on a 9×9 board

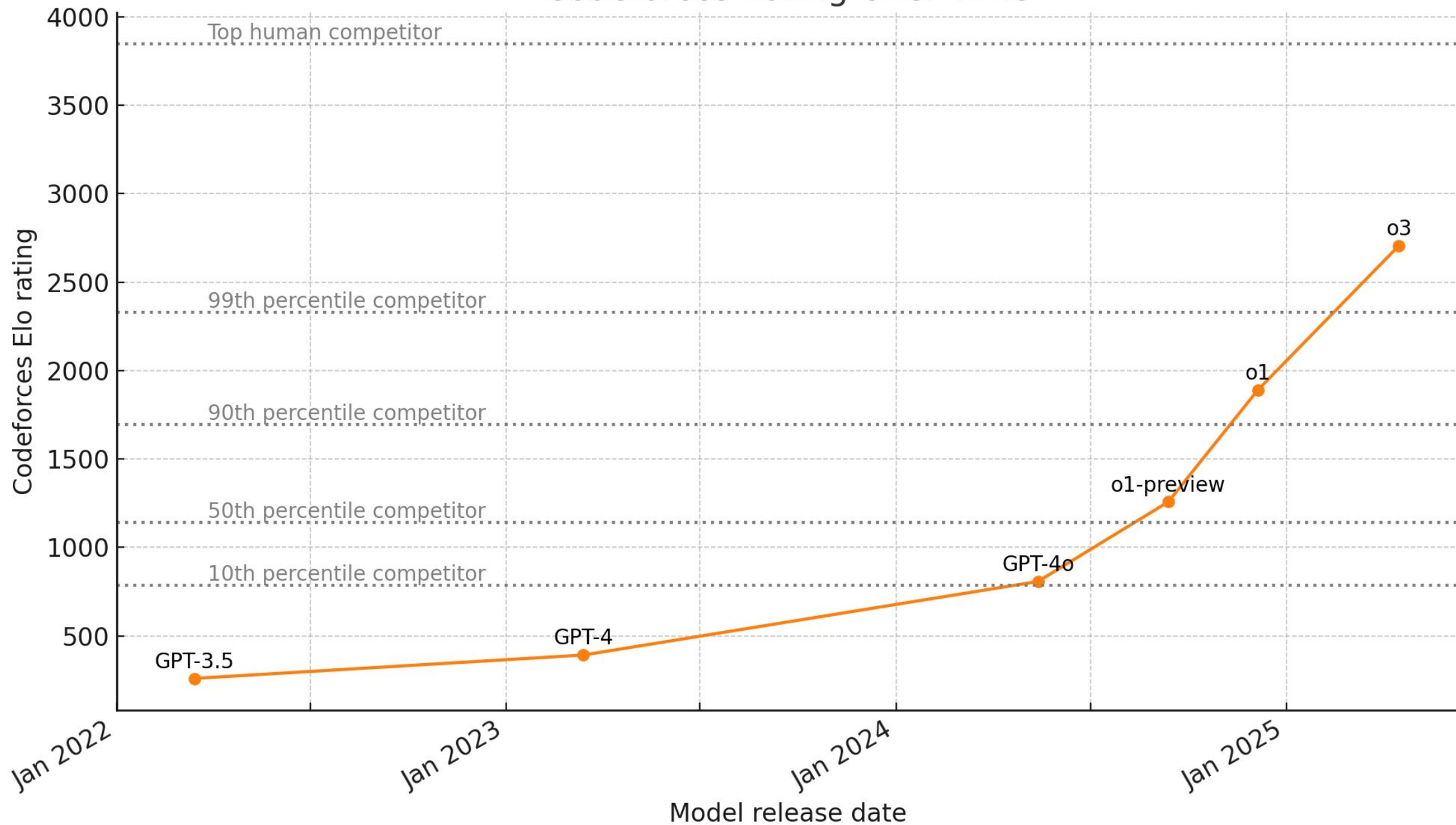
OpenAI o1



OpenAI o1 → o3



Codeforces Rating Over Time



https://github.com/noambrown/poker_solver

Node **b250**

Fold (blue) • Check/Call (green) • Bet/Raise (red) • Height = reach

	A	K	Q	J	T	9	8	7	6	5	4	3	2
A	AA	AKs	AQs	AJs	ATs	A9s	A8s	A7s	A6s	A5s	A4s	A3s	A2s
K	AKo	KK	KQs	KJs	KTs	K9s	K8s	K7s	K6s	K5s	K4s	K3s	K2s
Q	AQo	KQo	QQ	QJs	QTs	Q9s	Q8s	Q7s	Q6s	Q5s	Q4s	Q3s	Q2s
J	AJo	KJo	QJo	JJ	JTs	J9s	J8s	J7s	J6s	J5s	J4s	J3s	J2s
T	ATo	KTo	QTo	JTo	TT	T9s	T8s	T7s	T6s	T5s	T4s	T3s	T2s
9	A9o	K9o	Q9o	J9o	T9o	99	98s	97s	96s	95s	94s	93s	92s
8	A8o	K8o	Q8o	J8o	T8o	98o	88	87s	86s	85s	84s	83s	82s
7	A7o	K7o	Q7o	J7o	T7o	97o	87o	77	76s	75s	74s	73s	72s
6	A6o	K6o	Q6o	J6o	T6o	96o	86o	76o	66	65s	64s	63s	62s
5	A5o	K5o	Q5o	J5o	T5o	95o	85o	75o	65o	55	54s	53s	52s
4	A4o	K4o	Q4o	J4o	T4o	94o	84o	74o	64o	54o	44	43s	42s
3	A3o	K3o	Q3o	J3o	T3o	93o	83o	73o	63o	53o	43o	33	32s
2	A2o	K2o	Q2o	J2o	T2o	92o	82o	72o	62o	52o	42o	32o	22

Details

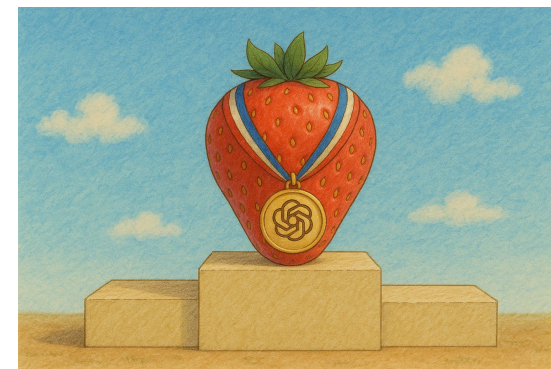
AKs @ b250

Reach (class): 72.07%

Mix: fold 84.4% • check/call 15.5% • bet/raise 0.1%

Hand	Reach	Actions		
KsAs	61.3%	Fold	96.3% Call	3.7% Rais
KhAh	75.6%	Fold	51.6% Call	48.0% Rais
KdAd	90.7%	Fold	95.7% Call	4.3% Rais
KcAc	60.7%	Fold	96.4% Call	3.6% Rais

OpenAI IMO Gold Medal Performance



- 5/6 problems correct (gold medal performance)

- Same constraints as human contestants:

- Within the 4.5 hour time limit
- No tools/browser/RAG
- Read/Write in natural language

- Each problem graded by 3 external former IMO medalists

- Proofs released publicly for anyone to vet:

<https://github.com/aw31/openai-imo-2025-proofs/>

PROBLEM 1 SUBMISSION

```
\[
P_n:=\{(a,b)\in \mathbf{Z}_{>0}^2: a+b\leq n+1\}, \quad n\geq 3.
\]
```

Need families of n lines covering P_n . "Sunny": direction not in {horizontal, vertical, slope -1}.
Need possible numbers of sunny lines in an n -line covering.

1. Basic configs, notation.

For n : triangle in plane: $T_n = \{x=1, y=1, x+y\leq n+1\}$, vertices $(1,1), (1,n), (n,1)$. P_n = integer poi

Three sides lines: H_n : $y=1$ bottom, V_n : $x=1$ left, D_n : $x+y=n+1$ hyp. Exactly forbidden directions.

So "non-sunny" = line parallel to one of triangle sides. Good.

Let $K_n = \{k : \text{exists } n \text{ distinct lines covering } P_n \text{ with exactly } k \text{ sunny}\}$.

$n\geq 3$ fixed.

Goal: determine K_n for all n .

Will prove: $K_n=\{0,1,3\}$, independent of n .

So main is: (a) for $n\geq 4$, any n -line cover must contain a side-line; inductively reduce to $n=3$. (b)

Also need constructions for $k=0,1,3$ for all n .

2. Lemma: $n\geq 4$, any n -line cover of P_n must use a side of triangle.

Call boundary point set of triangle:

$B_n = P_n \cap (H_n \cup V_n \cup D_n)$: all integer points on 3 sides.

Implications: Toward Scientific Discovery

- GSM8K problems take a human mathematician ~ 0.1 minutes
 - Saturated in 2023
- MATH problems take a human mathematician ~ 1 minute
 - Saturated in 2024
- AIME problems take a human mathematician ~ 10 minutes
 - Saturated in 2025
- IMO problems take a human mathematician ~ 100 minutes

Implications: Toward Scientific Discovery

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- IMO problems take a human mathematician ~100 minutes
- Real research takes a person **weeks to years**

Implications: Toward Scientific Discovery

- Frontier reasoning LLMs are starting to meaningfully contribute to scientific research
- I expect this to become more and more common over the next couple years

Title: New Results on Quantum Oracles

Speaker: Scott Aaronson

Abstract: I'll describe two new results that both involve quantum oracles and low-degree rational approximations of the sign function (and were both proved with some help from GPT-5). First, every black-box amplification protocol for the class QMA, or Quantum Merlin-Arthur, must yield completeness error at least $1/\exp(\exp(n))$. This precisely matches a striking recent protocol by Stacey Jeffery and Frek Witteveen (arXiv:2506.15551). Second, $PQP = \text{PostBQP}$ relative to all quantum oracles closed under either inverses or complex conjugation, but not all quantum oracles lacking those properties. Along with recent work by Ewin Tang and John Wright (arXiv:2507.23787) among others, this demonstrates the importance of the closure properties of quantum oracles.

Where does this go?

- There is still room to push inference compute **much further**
- Much **higher inference compute**, but much **more capable models**
 - What inference cost are you willing to pay for a proof of the Riemann Hypothesis?
 - What inference cost are you willing to pay for new life-saving drugs?

The Bitter Lesson by Richard Sutton

“The biggest lesson that can be read from 70 years of AI research is that general methods that leverage computation are ultimately the most effective... The two methods that seem to scale arbitrarily in this way are *search* and *learning*.”

Thank You!

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Lunch 🤪

Leave any type of feedback at pkr.bot/feedback!



Closing Remarks
:D

The background is a solid blue color. On the left side, there are several overlapping circles of varying shades of blue, creating a decorative, abstract pattern.

Giveaway Winner!

Thanks for watching!

Information hub at pkr.bot/hub

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